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Next Generation Water Prediction Capability (NGWPC)

NGWPC ngenCERF Users Guide

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A	Refer to cPDM for release date	Calibration and Evaluation Sections	Updates reflecting formulation selection release.
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1.0 Introduction

The Next Generation (**ngen**) tool for **Calibration**, **Evaluation**, **Reforecast**, and **Forecast** (ngenCERF) graphical user interface (GUI) supports four primary workflows: **Calibration**, **Evaluation**, **Forecast**, and **Verification**. Reforecast is also referred to as verification or hindcasting. The **ngenCERF** tool may be operated either through the GUI or via the Command Line Interface (CLI).

Originally, ngenCERF relied on the ngen-cal¹ module for calibration and validation; however, this module has since been deprecated within the NGWPC program. The tool has been refactored into several modular components, including a **Calibration Manager**, **Evaluation Manager**, **Model Setup Workflow Manager**, and **Forecast Manager**.

During calibration, the Calibration Manager repeatedly executes the ngen program with a user-selected formulation of hydrologic models for a specified gage, incorporating forcing data and iteratively adjusted parameter sets. After each ngen run, outputs are evaluated against historical observational data, and calibration parameters are automatically updated prior to the next iteration. This process continues until the defined stopping criteria is met.

Upon completion of calibration, a final validation run is performed using the best parameter set identified during the calibration process. The resulting outputs are available through the Evaluation workflow and may be downloaded to the user's local system.

The Forecast workflow uses the best parameter set produced during calibration to perform a user-selected forecast. ngenCERF currently supports eight distinct forecast types, allowing users to tailor forecast execution to their specific modeling objectives. Each forecast run is retained and may be individually selected for further analysis.

Following the completion of a forecast, users may run a verification on a selected forecast run in the Verification workflow. Verification compares forecast outputs against corresponding observational or reference datasets, enabling users to assess forecast performance for a specific forecast configuration.

All times referenced in this guide, unless otherwise noted, are in Coordinated Universal Time (UTC).

2.0 Definitions

The following table provides definitions to terms used in the ngenCERF tool and is not exhaustive.

Table 1: Definitions of Terms

Term	Definition
Catchment	A catchment is a hydrology-specific control volume with specific characteristics: a realized flowpath connecting inflow (or headwater) to outflow; and an area bounded by a divide that connects landscape processes to a flow network.

¹ ngen-cal was originally developed by NOAA OWP and updated for NGWPC.

Term	Definition
Domain	Geographical area of interest. E.g. CONUS, Alaska, Hawaii, Puerto Rico/US Virgin Islands.
Forcing data	Hydrological and atmospheric forces on a catchment, such as amount of rain, wind velocity, etc.
Formulation	Selection of modules defining the model stack to run in ngen.
Gage	Measures stream flow. For the purpose of calibration, it is also known as the Headwater Basin Gage.
GeoPackage	The hydrofabric file associated with a specific gage. This is often called Subsetted Hydrofabric.
Hydrofabric	A dataset containing a network of connected representations of rivers, lakes, and catchments.
Data Services	Provides API calls to retrieve forcing, observational, subsetted hydrofabric GeoPackage, and module initial parameters based upon a Gage ID and formulation. This name for this API is still evolving so this document uses the generic term Data Services to refer to data provided by the NGWPC Enterprise Data and FIM Services (EDFS) team.
Metrics	Statistics calculated by the calibration manager during calibration.
Objective Function	Metric used as the error function on the output variable.
Observational data	Historical streamflow data observed at a gage.
Optimization Algorithm	Algorithm used to optimize the calibration.
Reference dataset	Output from NWM v3.0.
Subsetted Hydrofabric	The hydrofabric associated with a specific gage. This is also called the GeoPackage.
Tuning Controls	Times, calibratable parameters, and output variable for performing a calibration with ngen-cal.

3.0 Running ngenCERF

ngenCERF may be executed using either the **Command Line Interface (CLI)** or the web-based **Graphical User Interface (GUI)**. To use either interface, users must first log in to **Parallel Works (PW)** and connect to the shared operational environment (OE) session.

The information required to perform a calibration is stored in a combination of database tables and flat files, including subsetted datasets retrieved from Data Services. The GUI guides users through the calibration setup process using a sequence of tabs designed to collect or retrieve the necessary inputs. When all required information has been provided, the job status is set to **Ready**, allowing the calibration to be executed.

When running a calibration from the CLI, a **Calibration Definition File** is required. This file may be exported from a previously completed calibration job and modified as needed to support new calibration runs.

4.0 ngenCERF CLI

The ngenCERF Command Line Interface (CLI) provides a text-based interface for interacting with the ngenCERF server, enabling users to create, update, manage, and execute calibration jobs. The CLI serves as an alternative to the GUI and supports automation and scripting workflows. This section describes the commands available through the CLI. For the most up-to-date command information, refer to the [ngenCERF CLI Confluence page](#).

4.1 When to Use the CLI

The ngenCERF CLI is intended for users who already have an existing calibration configuration file and wish to execute a calibration without launching the GUI. It is particularly well suited for automated workflows, such as submitting a large batch of calibration jobs for execution or integrating calibration runs into scripted or scheduled processes.

Regardless of how a calibration is submitted, all calibration output is accessed through the GUI. The GUI provides tools to monitor calibration progress, evaluate results, and download output data to the user's local system.

4.2 How to Use the CLI

To access the CLI, log in to **Parallel Works**, select **Clusters** from the left navigation panel, and double-click the **OE cluster** (use UAT Cluster during the delivery check-out phase). In the upper-right corner of the cluster view, your username and associated IP address are displayed. Click on your username to open a terminal session, where you can run ngenCERF CLI commands.

4.3 CLI Commands

All CLI commands are invoked using the **ngencertif**, followed by a command name and any required or optional arguments, as defined by the command's parameters. All times referenced are in Coordinated Universal Time (UTC).

Syntax

ngencerf <command name> <args>

Legend

< > Angle Brackets indicate required information
[] Square brackets indicate an optional argument

4.3.1 about

Returns the commit hash and commit details for each ngenCERF repository associated with the calibration job.

Usage

ngencerf about [--output <location>]

Arguments

--output, -o Optional output directory for the file containing the exported information (default: current working directory).

Example

ngencerf about --output ~/ngencerf

4.3.2 archive

Archives one or more calibration jobs, removing them from most operations without permanently deleting them.

Usage

ngencerf archive <run_ids | file_path>

Arguments

run_ids A space-separated list of one or more calibration run IDs.
file_path A file containing the output of the **ngenerf jobs** command, listing the jobs to be deleted

Example

ngencerf archive 123

OR

ngencerf archive 123 145 201

4.3.3 cancel

This command will cancel a running calibration job id.

Usage

ngencerf cancel <run_id>

Arguments

run_id Required. Integer number.

Example

```
ngincerf cancel 123
```

4.3.4 delete

Permanently deletes one or more calibration jobs after prompting for confirmation.

Usage

```
ngincerf delete <run_ids | file_path>
```

Arguments

run_ids A space-separated list of one or more calibration run IDs to delete.
file_path A file containing the output of the **ngincerf jobs** command, listing the jobs to be deleted

Example

```
ngincerf delete 123
```

OR

```
ngincerf delete 123 145 201
```

4.3.5 download

Downloads a ZIP archive for a calibration run.

Usage

```
ngincerf download <run_id> [--output OUTPUT]
```

Arguments

run_ids A space-separated list of one or more calibration run IDs to delete.
--output, -o Optional output directory for the file containing the exported information (default: current working directory).
working directory.

Example

```
ngincerf download 123
```

OR

```
ngincerf download 123 --output ~/Downloads/job_123.zi
```

4.3.6 export

Exports an existing calibration from the ngenCERF tool. The resulting Calibration Definition File can be used with the **ngincerf import** command or as a template for creating new calibrations. See the table below for descriptions of the Calibration Definition File entries and refer to the Data section for an example JSON file. By default, the exported file is named using the job ID and username.

Usage

```
ngincerf export <run_id> [--output OUTPUT] [--show]
```

Arguments

run_id	Required. Integer number.
--output, -o	Optional output directory for the file containing the exported information (default: current working directory).
--show, -s	Optional to display the job in the console.

Example

```
ngincerf export 123 --output ./job_456.json --show
```

4.3.7 import

Imports a new calibration job from a **Calibration Definition File** (JSON), such as one generated by the `ngincerf export` command. The `--run` flag may be specified to start the job immediately; if provided, it overrides the `run_after_import` setting in the JSON file.

Usage

```
ngincerf import input_file [--run true|false]
```

Arguments

input_file	The path to the JSON file containing the Calibration definition file. See the table below and the Data section for an example JSON file.
--run, -r	(Optional) Overrides the <code>run_after_import</code> field in the JSON file. Defaults to true if specified without a value.

Example

```
ngincerf import my_job.json --run false
```

4.3.8 jobs

Lists all calibration jobs, optionally saving the output to a markdown file.

Usage

```
ngincerf jobs [--output OUTPUT]
               [--filters FILTERS][--sort SORT]
               [--gage-id GAGE_ID]
               [--status STATUS [STATUS ...]]
               [--include-archived true|false]
               [--module-operator and|or]
               [--module-list MODULE [MODULE ...]]
               [--date-operator before|after|between]
               [--date DATE | --date-start DATE --date-end DATE]
               [--id-operator before|after|between]
               [--id ID | --id-start ID --id-end ID]
```

```
[--sort-field FIELD]
[--sort-direction asc|desc]
```

Arguments

All of these arguments are *OPTIONAL*:

--output, -o	Output directory for the file containing the exported information (default: current working directory).
--filters	JSON or YAML file path, or inline JSON/YAML string defining filters. Examples: <pre>--filters filters.yaml</pre> <pre>--filters '{"gage_id": "01544887", "status": ["Done", "Failed"]}'</pre>
--sort	JSON or YAML file path, or inline JSON/YAML string defining sort order. Examples: <pre>--sort sort.yaml</pre> <pre>--sort '{"field": "submit_date", "direction": "desc"}'</pre>
--gage-id	Filter jobs by exact gage ID.
--status	Filter by one or more statuses (e.g., Done Failed Running).
--include-archived	Include archived jobs. Default: false.
--module-operator	Combine multiple modules using and (default) or or .
--module-list	List of module names to match (e.g., CFE-X Noah-OWP-Modular)
--date-operator	One of before , after or between
--date	Date in YYYY-MM-DD format (for before or after)
--date-start/--date-end	Start and end dates (for between)
--id-operator	One of before , after or between
--id	Job ID (used with before or after).
--id-start/id-end	Range of job IDs (used with between).
--sort-field	Field to sort by. Options: ○ gage_id ○ user_formulation_name ○ submit_date ○ create_date ○ job_genesis ○ status ○ calibration_start_period ○ calibration_end_period ○ stop_criteria
--sort-directory	Sorting direction (asc or desc (default))

Example

```
ngencerf jobs --output ./all_jobs.md
or
ngencerf jobs \
  --gage-id 01055000 \
  --status Done Failed \
  --module-operator or \
```

```
--module-list "CFE-X" "Noah-OWP-Modular" \  
--date-operator after \  
--date 2025-10-10 \  
--sort-field submit_date \  
--sort-direction desc \  
--output ./recent_jobs.md
```

4.3.9 lock

Locks one or more calibration jobs, preventing them from being deleted or archived.

Usage

```
ngincerf lock <run_ids | file_path>
```

Arguments

run_ids	A space-separated list of one or more calibration run IDs to delete.
file_path	A file containing the output of the ngincerf jobs command, listing the jobs to be deleted

Example

```
ngincerf lock 1 2 3  
or  
ngincerf lock calibration_jobs_2025-11-04_1455.md
```

4.3.10 register

Registers a new user for the ngenCERF server. The credentials are saved in **~/.ngincerf_env** for future CLI sessions.

Usage

```
ngincerf register [email]
```

Arguments

email	Optional email address to register. If omitted, the CLI will prompt for it..
--------------	--

Example

```
ngincerf register user@example.com
```

4.3.11 run

Runs an existing calibration job by job ID, if it has not yet been executed or is not currently running. This command is only required if the **run_after_import** option was set to **false** during **ngincerf import**, or if the calibration job was created in the GUI but has not yet been run.

Usage

```
ngincerf run <run_id>
```

Arguments

run_id	Required. Integer number.
---------------	---------------------------

Example

```
ngencerf register user@example.com
```

4.3.12 show

Displays the details of a calibration job.

Usage

```
ngencerf show <run_id>
```

Arguments

run_id	The calibration run ID to display.
---------------	------------------------------------

Example

```
ngencerf show 123
```

4.3.13 unarchive

Unarchives one or more calibration jobs, restoring them to active status.

Usage

```
ngencerf unarchive <run_ids | file_path>
```

Arguments

run_ids	A space-separated list of one or more calibration run IDs to delete.
file_path	A file containing the output of the ngencerf jobs command, listing the jobs to be unarchived.

Example

```
ngencerf unarchive 1 2 3
```

or

```
ngencerf unarchive calibration_jobs_2025-11-04_1455.md
```

4.3.14 unlock

Unlocks one or more calibration jobs, allowing them to be deleted or archived.

Usage

```
ngencerf unarchive <run_ids | file_path>
```

Arguments

run_ids	A space-separated list of one or more calibration run IDs to delete.
file_path	A file containing the output of the ngencerf jobs command, listing the jobs to be unlocked.

Example

```
ngencerf unlock 1 2 3
```

or
ngincerf unlock calibration_jobs_2025-11-04_1455.md

4.3.15 update

Updates an existing calibration job that has not been run with new parameters from a JSON file.

Usage

```
ngincerf update <run_id> <input_file>
```

Arguments

run_ids	Calibration run ID to update
input_file	The path to the JSON file containing the updated job parameters

Example

```
ngincerf update 456 updated_job.json
```

4.4 Output Files

Several subcommands save output files to disk. By default, files are written to the current working directory, with command-specific default filenames.

The default location and/or filename can be overridden using the optional **--output** or **--export** argument (depending on the command). The behavior is as follows:

- **Directory only** – Specify a directory (e.g., **--output ~/foo/**). The default filename is used within that directory.
- **Filename only** – Specify a filename without a directory (e.g., **--output new-file.txt**). The file is saved to **current working directory** with the given name.
- **Full path** – Specify a full path (e.g., **--output ~/foo/new-file.txt**) The file is saved exactly at that location

Examples:

- | | |
|--------------------------------------|--|
| • --output ~/ | saves to ~/my-default.txt |
| • --output ~/foo/ | saves to ~/foo/my-default.txt |
| • --output new-file.txt | saves to <current working directory>/new-file.txt |
| • --output ~/foo/new-file.txt | saves to ~/foo/new-file.txt |

4.5 Importing and Exporting

4.5.1 Calibration Definition File

The **Calibration Definition File** is a JSON file containing all entries required to fully set up and execute a calibration. It can be generated from an existing calibration job and used either directly as input for a new job or as a template for creating a custom calibration setup.

The metadata section of this file is ignored for the **ngencerf import** command. Users may include information in this section for reference or documentation purposes.

See **Section 6.5** for an example JSON File.

Table 2: Calibration Definition File

Entry	Description
source_calibration_run_id	Metadata: ID of the job that was exported.
time_range	Metadata: Intersection of time range from the forcing data and observation data, constraining calibration and validation.
errors	Metadata: Shows any outstanding errors in the job that would prevent it from running.
run_after_import	Flag indicating whether to submit the job immediately after import provided all required information is available and there are no errors. Can be overridden by the --run option of import or update .
gage_id	The gage_id associated with the calibration job.
forcing_source	The source of the forcing data, e.g., AORC or NWM Retrospective.
observational_source	The source of the observational data, e.g., Historical.
geopackage_source	The source of the geopackage data e.g., Hydrofabric
modules	The list of modules used for this calibration.
formulation_name	User-supplied formulation name that has no effect on the job .
use_sloth	Flag indicating the SLoTH model will be added to the formulation. . If true, sloth_parameters must be provided. Note: The current calibration manager design does not allow for specification of SLoTH variables.
sloth_parameters	Required if use_sloth is true. The SLoTH parameter definitions. All SLoTH fields are required for each SLoTH variable. Note: The current calibration manager design

Entry	Description
	<i>does not allow for specification of SLoTH variables.</i>
automatic_validation	If true, a validation run is executed automatically after the calibration run. <i>Note: Due to the current calibration manager design, this will always true.</i>
calibration_times	The time ranges to use for calibration.
validation_times	The time ranges to use for validation (Must be outside the calibration times).
stream_flow_threshold	Required whenever a categorical metric is to be calculated. If this field is omitted or null and the categorical metric is not selected as the objective function, the Calibration Manager will skip calculating the categorical metric. .
peak_flow_threshold	Required whenever an event-based metric is to be calculated. If this field is omitted or null and the event-based metric is not selected as the objective function, the Calibration Manager will skip calculating the event-based metric.
parameters	Defines the module parameters used for calibration tuning. Each entry must include name, min, max, and module.
objective_function	Metric to use for the objective function.
optimization	Optimization algorithm (DDS, GWO, PSO).
optimization_inputs	Inputs for the selected optimization.
save_plot_iteration_frequency	How often the calibration manager will generate plots during the calibration.
save_output_iteration	If true, output for each iteration is saved in separate files (not supported by UI).
stop_criteria	Number of worker iterations to run.

5.0 ngenCERF GUI

The GUI consists of a series of pages with tabbed work areas for configuring and running the **Calibration**, **Validation**, **Forecast**, and **Verification** workflows (also known as reforecasting or hindcasting). To access the GUI, log in to **Parallel Works** (PW) and connect to the shared operational environment (OE) session. Once connected, the ngenCERF Login Page is displayed. For better performance and troubleshooting, it is recommended to use the **Open in Browser** button in the upper-right corner to launch the GUI, which opens the login page directly in your browser.

Contextual help is provided throughout the GUI, specific to each page or tabbed work area. Within the **Calibration** workflow, a summary panel on the right side of the page provides an overview of the calibration setup, including progress indicators and details of the current job configuration

5.1 Login Page

Enter your ngenCERF username and password or request a ngenCERF account.

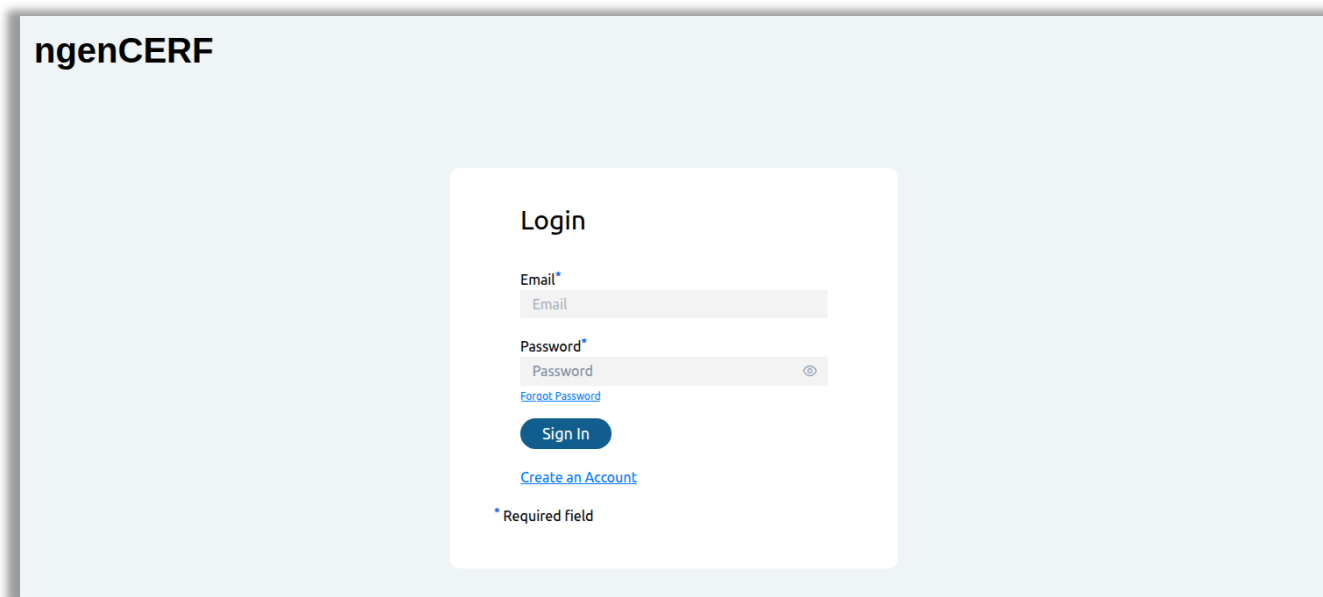
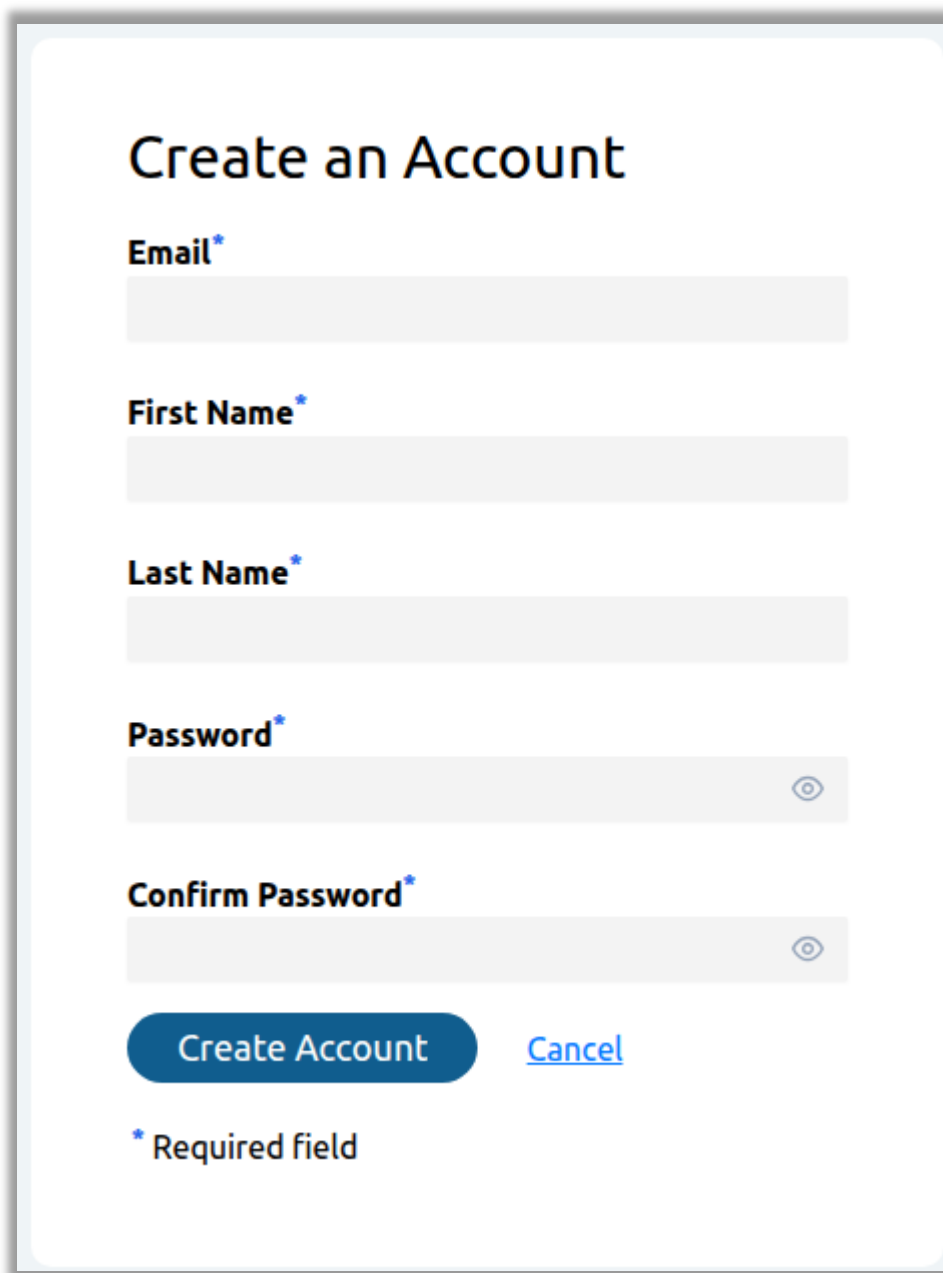
The screenshot shows the ngenCERF Login page. The page has a light blue background. In the top left corner, the text "ngenCERF" is displayed. Centered on the page is a white login form. The form has a title "Login". Below the title are two input fields: "Email*" and "Password*". The "Email*" field has a placeholder text "Email". The "Password*" field has a placeholder text "Password" and a toggle icon (an eye) to its right. Below the "Password*" field is a link "Forgot Password". Below the "Forgot Password" link is a blue button labeled "Sign In". Below the "Sign In" button is a link "Create an Account". At the bottom of the form, there is a small text "* Required field".

Figure 1 Login Screen

To create an account, provide a username, password, and email.



The image shows a 'Create an Account' form with the following fields and elements:

- Create an Account** (Section Header)
- Email*** (Text label above a text input field)
- First Name*** (Text label above a text input field)
- Last Name*** (Text label above a text input field)
- Password*** (Text label above a password input field with an eye icon for visibility toggle)
- Confirm Password*** (Text label above a password input field with an eye icon for visibility toggle)
- Create Account** (Blue button)
- [Cancel](#) (Blue text link)
- * Required field** (Footnote text)

Figure 2 Create Account

To log out, click the User Menu (your initials) in the upper-right corner of the application and select Logout .

5.2 Landing Page

The ngenCERF landing page serves as the primary entry point to the application. Across the top of the page is the standard workflow navigation bar, which includes the ngenCERF logo, workflow buttons for **Calibration**, **Evaluation**, **Forecast**, and **Verification**, as well as access to **Help** and the **User Menu**.

Upon logging in, users are greeted with a welcome message and a dashboard that provides a high-level summary of calibration activity. The dashboard displays three informational cards showing the number of calibrations that are currently **Running**, **Ready to Run**, and **In Setup**. These indicators provide quick insight into the current state of calibration jobs.

From the landing page, users select one of the available workflows to proceed with their tasks.

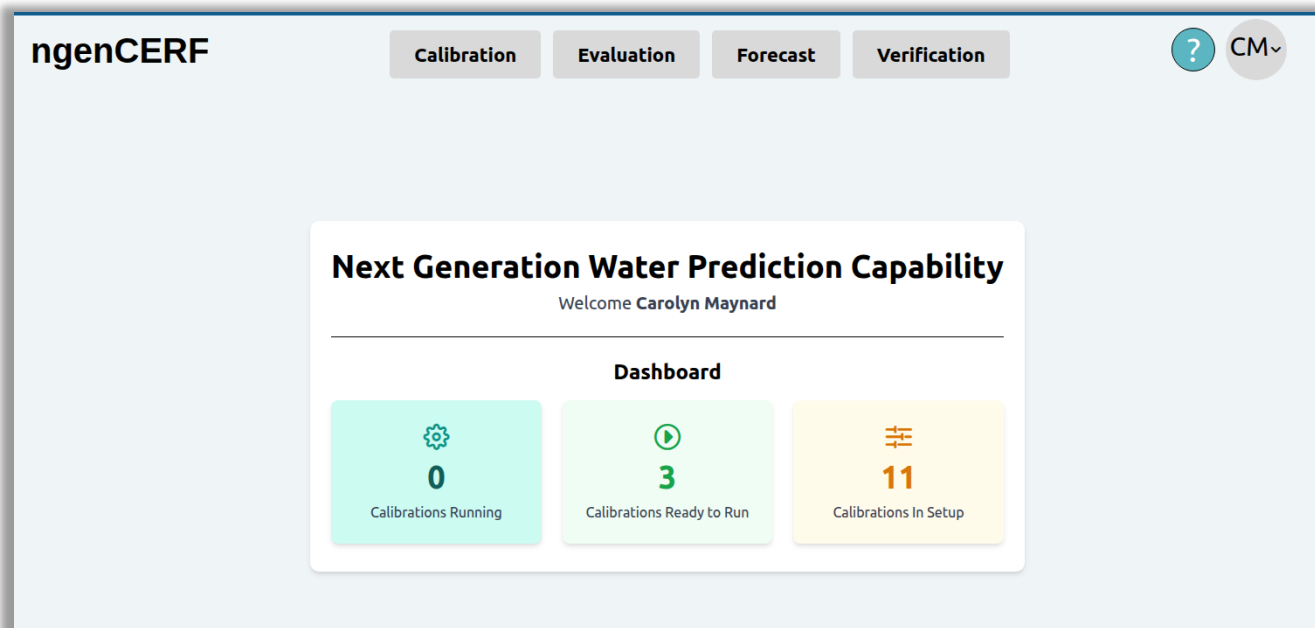


Figure 3 Landing Page

Header Workflow Buttons:

Calibration	Opens the Calibration workflow.
Evaluation	Opens the Evaluation workflow. Available only after at least one calibration job has completed successfully.
Forecast	Opens the Evaluation workflow. Available only after at least one calibration job has completed successfully.
Verification	Opens the Verification workflow. Available only after at least one forecast job has completed successfully.

5.3 ngenCERF Interface Basics

ngenCERF is organized into four primary workflows: Calibration, Evaluation, Forecast, and Verification. Selecting a workflow displays its corresponding work area, which is organized into a series of tabs specific to that workflow.

Contextual help is available throughout the application by selecting the question mark icon in the header area. In addition, right-click context menus provide access to actions associated with calibration jobs displayed in tables.

5.3.1 ngenCERF Header Area

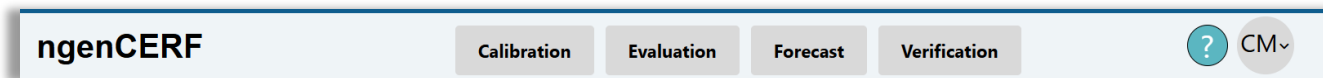


Figure 4 ngenCERF Header Area

The header area displays the ngenCERF logo, workflow buttons, the Help button, and the user menu. The Help button provides contextual assistance based on the selected workflow and tab. Click the user's initials to open the **User Menu**, which provides access to **About**, **Account**, **Notifications**, this **Users Guide**, and **Logout**.

User Menu:

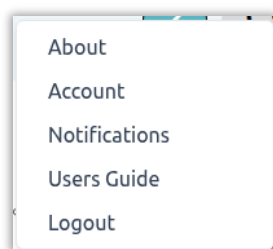


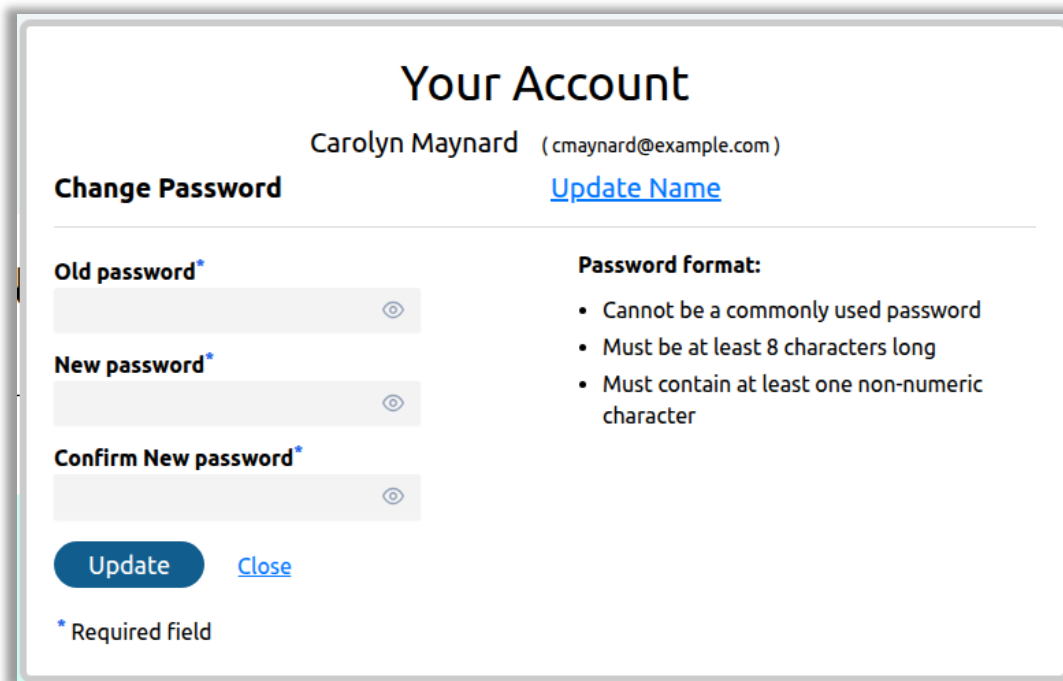
Figure 5 Header User Menu

Options:

About	Displays table with the version release tag or commit hash for all repositories associated with the ngenCERF release.
Account	Opens User Profile page to update user details, including password.
Notifications	Provides a table of all UI notification messages generated during the users ngenCERF session.
Users Guide	Opens this Users Guide in a separate browser tab for viewing and downloading.
Logout	Logs the user out of ngenCERF.

5.3.2 User Profile Page

Use this page to update your username and password.



Your Account

Carolyn Maynard (cmaynard@example.com)

Change Password

[Update Name](#)

Old password*

New password*

Confirm New password*

Update

[Close](#)

* Required field

Figure 6 User Profile

5.4 Calibration Workflow

The Calibration workflow enables users to configure and run an automated calibration, followed by an automatic validation run using the best parameters identified during calibration. Calibration metrics are computed at each iteration and are available for review both while the job is running and after it completes.

During calibration, ngen is executed iteratively using updated parameter sets. Each iteration adjusts parameters based on the results of the previous run to produce simulated values that match the observed output variable as closely as possible. Currently, streamflow is the only supported output variable, as it is the only observed data available to the calibration manager.

After the calibration run successfully completes, a validation run is executed automatically. This validation consists of a single ngen run over a time period different from the calibration period, using the best parameters determined during calibration.

All times referenced are in Coordinated Universal Time (UTC).

Calibration Workflow Interface

The Calibration Workflow Interface provides the tools required to configure all aspects of a calibration, including:

- Selection of the Headwater Basin Gage

- Selection of hydrologic modules and naming of the formulation (also referred to as the Model Stack)
- Calibration and Validation time periods
- Output Variable to Calibrate against (currently streamflow)
- Calibration Tuning Parameters
- Optimization Algorithm and Objective Function
- Additional Metrics
- Calibration Stop criteria
- Plot generation frequency
- Error, Warning, and Trapping System (EWTS) settings

All necessary data—including forcing data, observational data, and the subsetting hydrofabric GeoPackage are obtained automatically based on the selected gage. The set of available calibratable parameters is retrieved from Data Services based on the selected model formulation. Users may select parameters from this list for calibration tuning or upload a parameter set from a file.

Calibration Workflow Tabs

The Calibration workflow is organized into the following tabs:

- **Calibration Runs** – Lists the user's calibration jobs
- **Headwater Basin Gage** – Displays the headwater basin gage, forcing data, and observational data
- **Formulation** – Select the model stack and provide a name for the formulation
- **Tuning Controls** – Times, Output Variable to Calibrate, and Calibration Tuning Parameters
- **Optimization / Metrics** – Optimization Algorithm, Objective Function, Additional Metrics, Stop Criteria, and Plot Generation Frequency
- **Run / Status** – Review the calibration setup, start the run, monitor progress, and view results

Users navigate the tabs using the **Next** and **Prev** buttons or by clicking a tab name.

Table 3: Available Calibration Setup Items

Item	Description	Tab Location
Domain	Select the domain of interest to narrow the list of gages.	Headwater Basin Gage
Gage	Select the Headwater Basin Gage ID.	Headwater Basin Gage
Forcing Data	Source for AORC data.	Headwater Basin Gage
Observational Data	Source for observational data (Historical only)	Headwater Basin Gage
GeoPackage	Source for GeoPackage (Data Services only)	Headwater Basin Gage
Formulation Name	User defined name for this formulation.	Formulation
Formulation	Selection of modules for the calibration run.	Formulation
Calibration Time Controls	Set of four (4) times defining the length of the simulation and when to start and end the calibration activities within the simulation time.	Tuning Controls
Validation Time Controls	Set of four (4) times defining the length of the simulation and when to start and end the validation activities within the simulation time.	Tuning Controls

Item	Description	Tab Location
Output Variable To Calibrate ³	Currently ngen calibration only supports Streamflow.	Tuning Controls
Calibration Tuning Parameters ³	The set of calibratable parameters from the selected modules that the Calibration Manager adjusts to fine-tune the calibration.	Tuning Controls
Optimization Algorithm ³	The optimization algorithm to be used for the calibration.	Optimization / Metrics
Calibration Stop Criteria ³	The number of ngen iterations for each optimization worker .	Optimization / Metrics
Objective Function ³	A metric to use as the objective function.	Optimization / Metrics
Additional Metrics ³	Optional. Calculate optional Categorical or Event Based Metrics.	Optimization / Metrics
Plot Generation Frequency ³	Determines how often to generate plots during the calibration. Each plot is cumulative. One (1) is every iteration, five (5) is every five iterations, etc.	Optimization Metrics
Global Logging	Enable or Disable the ngen Error, Warning, and Trapping System Logging	Status / Run
Log Levels	Set the Error, Warning, and Trapping System log level for ngen, forcing, and each individual module in the formulation.	Status / Run

³ Not available in formulations with LSTM

The diagram below shows the typical workflow for setting up a calibration run. Trapezoids represent a Calibration GUI tab, with the items configured in that tab listed beneath it.

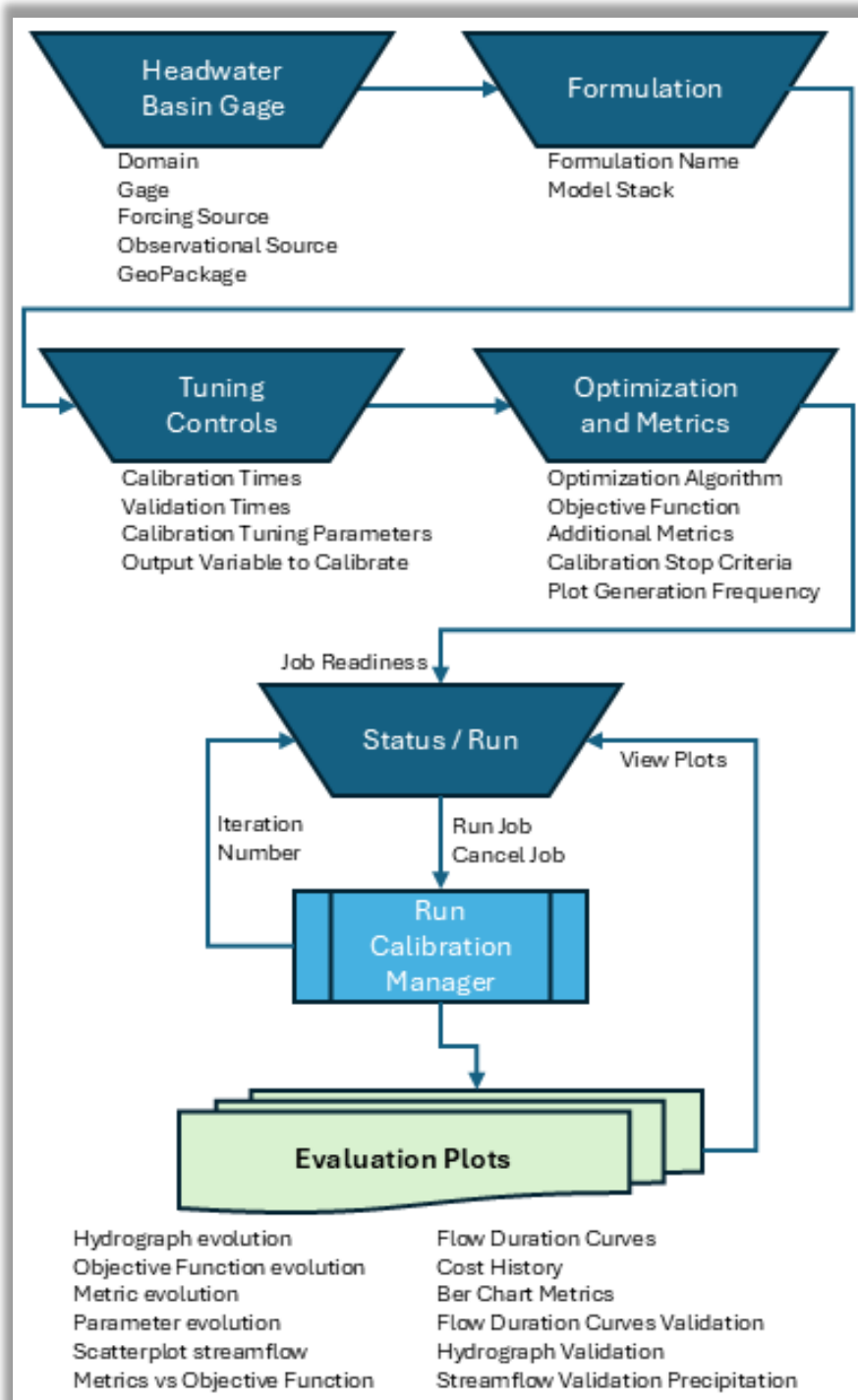
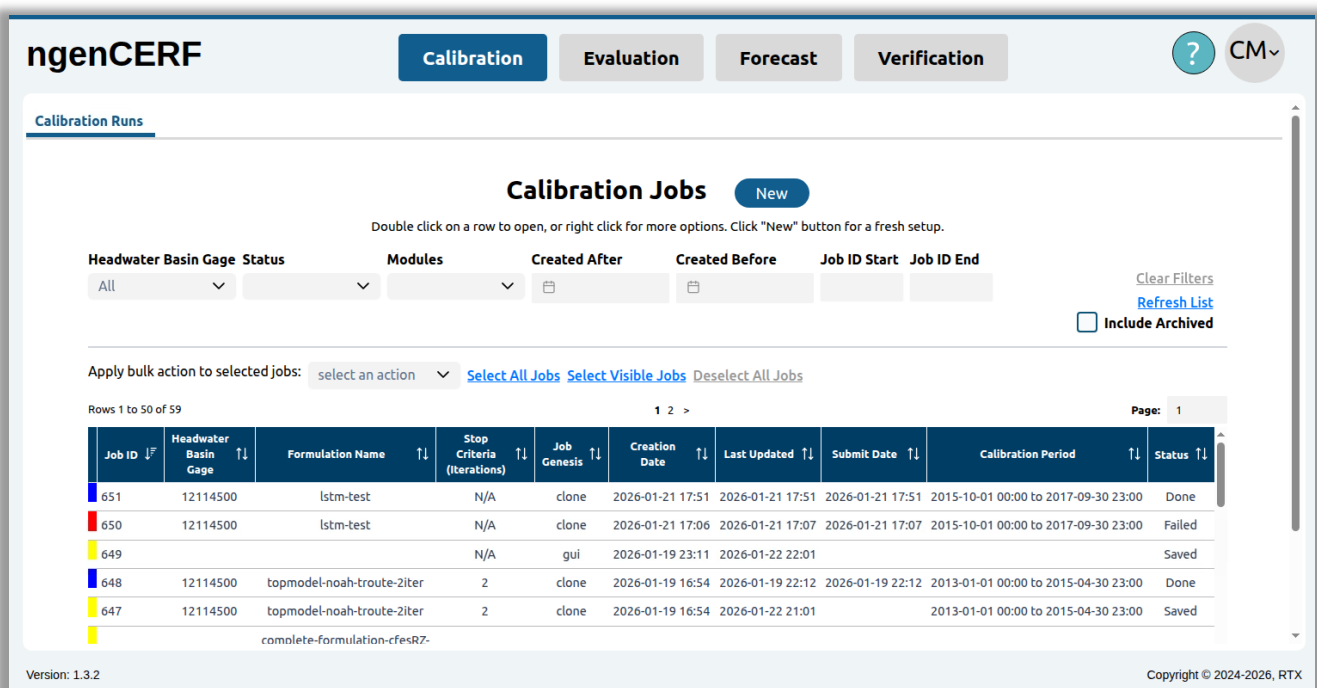


Figure 7: Typical Steps For Calibration Setup

5.4.1 Calibration Runs Tab

The **Calibration Runs** tab is the landing page when the **Calibration** workflow is selected. It provides a complete overview of all calibration jobs associated with the user, regardless of status. From this tab, users can create a new calibration from scratch, clone an existing calibration as a baseline for a new job, or perform actions on one or more jobs using the right-click context menu or bulk actions.

A table displays key details for each calibration job, with sortable columns and flexible filtering options based on Headwater Basin Gage, Module(s), Status, Archived status, creation date, and Job ID. The tab also includes controls to include archived jobs, refresh the list, and managing multiple jobs simultaneously. Selecting a job allows users to open it, view its status, or access additional actions such as exporting the calibration configuration or downloading results.



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Calibration Runs

Calibration Jobs New

Double click on a row to open, or right click for more options. Click "New" button for a fresh setup.

Headwater Basin Gage Status Modules Created After Created Before Job ID Start Job ID End

All [v] [v] [v] [v] [v] [v]

[Clear Filters](#) [Refresh List](#) ☐ Include Archived

Apply bulk action to selected jobs: select an action [Select All Jobs](#) [Select Visible Jobs](#) [Deselect All Jobs](#)

Rows 1 to 50 of 59 1 2 > Page: 1

Job ID	Headwater Basin Gage	Formulation Name	Stop Criteria (Iterations)	Job Genesis	Creation Date	Last Updated	Submit Date	Calibration Period	Status
651	12114500	lstm-test	N/A	clone	2026-01-21 17:51	2026-01-21 17:51	2026-01-21 17:51	2015-10-01 00:00 to 2017-09-30 23:00	Done
650	12114500	lstm-test	N/A	clone	2026-01-21 17:06	2026-01-21 17:07	2026-01-21 17:07	2015-10-01 00:00 to 2017-09-30 23:00	Failed
649			N/A	gui	2026-01-19 23:11	2026-01-22 22:01			Saved
648	12114500	topmodel-noah-troute-2lter	2	clone	2026-01-19 16:54	2026-01-19 22:12	2026-01-19 22:12	2013-01-01 00:00 to 2015-04-30 23:00	Done
647	12114500	topmodel-noah-troute-2lter	2	clone	2026-01-19 16:54	2026-01-22 21:01		2013-01-01 00:00 to 2015-04-30 23:00	Saved
		complete-formulation-cfesRZ-							

Version: 1.3.2 Copyright © 2024-2026, RTX

Figure 8 Calibration → Calibration Runs Tab

The user can double-click a job to open it (navigating to the Run / Status tab), **Ctrl+Click** to select or deselect multiple individual jobs, **Shift+Click** to select a range of jobs between the last selected job and the clicked job, and **right-click** to access available actions.

User Actions:

New

Select this button to create a calibration setup from scratch.

Filters

Use the filters at the top of the table to narrow down the list of calibration jobs. You can use any combination of filters; only jobs that match all active filters will be shown. The table updates automatically as you select filters.

Bulk Actions	Allows you to perform actions on multiple calibration jobs at once.
Double Click	Opens calibration job and takes user to the Status / Run tab.
Right Click	Open, Clone, Download, Delete, Archive, or Unarchive the selected calibration job in the Calibration Runs table. The Download option appears for Done jobs only. If the job is already archived, the Unarchive option will be available.
Open	Opens calibration job and takes the user to the Status/Run tab.
Clone	This will make a copy of the calibration job set up as a new calibration job for editing.
Download Results	Download all data associated with a finished calibration job.
Export Calibration Config	Export the Calibration Definition File. (Used by the CLI.)
Delete	Permanently deletes the selected calibration job(s) from the system.
Archive/Unarchive	Archive or Unarchive the selected calibration job(s). Archived jobs Will no longer appear in any Calibration Runs table unless the Archived filter is active.
Lock/Unlock	Lock/Unlock the selected calibration job(s). Locking a job prohibits deletion. Once locked the job cannot be deleted until unlocked.

Table 4: Calibration Job Status Descriptions

Status	Description
Saved	Some information has been saved, but the setup is not complete. Check the Progress checkmarks for missing items. Click on the item to go to the appropriate tab.
Ready	All the necessary information has been provided, and the job is ready to Run. Click the Run button to start the run.
Validating and Preparing Job Data	The backend is subsetting the data to the selected time range for the calibration/validation runs, running processes that validate/create the ngen realization file and creating the calibration manager input config file to submit the job for execution.
Submitted	Calibration has been submitted and waiting for resources from the cluster to begin execution.
Calibration Running	The calibration has begun. The iteration number reflects the last completed iteration run of ngen. Iteration 0 is the control run executed by the calibration manager for metric calculations. Logs are viewable once the calibration is in a running state. Calibration plots are viewable once iteration 1 is complete. Validation plots are viewable once Validation Best is done.
Calibration Done, Validation Control Running	The calibration has completed. The Calibration Stop Criteria has been reached.. The calibration manager is currently running the control run for validation metric calculations.
Calibration Done, Validation Best Running	Both the calibration run, and validation control run have completed successfully. The validation best is running ngen with the best set of parameters determined during the calibration phase.

Done	Calibration and Validation runs of ngen have successfully completed. Results are viewable in both the Calibration->Status/Run tab and the Evaluation workflow.
Calibration Failed	A failure occurred causing the Calibration Job to crash. Logs and Plots are viewable in the Calibration->Status/Run tab. If running from within the GUI, the ngen.log file will be automatically displayed in the UI. The user can also select to view the logs from the stdout/stdout runs of ngen and the calibration manager, if they exist. Logs and Plots are viewable in the Calibration->Status/Run tab.
Calibration Done, Validation Control Failed	A failure occurred causing the Validation Control Job to crash. Logs and Plots are viewable in the Calibration->Status/Run tab. If running from within the GUI, the ngen.log file will be automatically displayed in the UI. The user can also select to view the logs from the stdout/stdout runs of ngen and the calibration manager, if they exist.
Calibration Done, Validation Best Failed	A failure occurred causing the Validation Best Job to crash. Logs and Plots are viewable in the Calibration->Status/Run tab. If running from within the GUI, the ngen.log file will be automatically displayed in the UI. The user can also select to view the logs from the stdout/stdout runs of ngen and the calibration manager, if they exist.
Server Error	The server failed interrupting the Calibration Job either during calibration, or one of the validation runs.
Cancelled	Calibration Job was cancelled by the user.

Note: A Calibration Job in this table refers to both the calibration run and the validation run.

Table 5 Job Status Color Codes

Color	Job Status
 Yellow	Saved
 Green	Ready
 Flashing Green	Running
 Blue	Done
 Orange	Cancelled
 Red	Failed
 Black	Server Error

5.4.2 Calibration Progress Area

The Calibration workflow includes a progress panel displayed on the right side of the page that tracks progress toward completing a calibration setup. This panel appears after the user selects **New** to create a calibration or opens an existing calibration job from the **Calibration Runs** tab.

The progress panel is divided into two sections. The **Progress** section at the top lists the required calibration items, with a basin map displayed to the right showing the associated catchments. As the user progresses through the workflow tabs and saves the required information, checkmarks appear next to completed items in this section. Once all required items are marked complete, the calibration job enters the **Ready** state and can be run.

Each item in the Progress section is a hyperlink. Selecting an item navigates the user directly to the corresponding tab, where the information can be viewed and—if the job has not yet been run—updated and saved.

The **Setup** summary section at the bottom of the panel provides a concise overview of the calibration configuration, listing the key selections for each required item

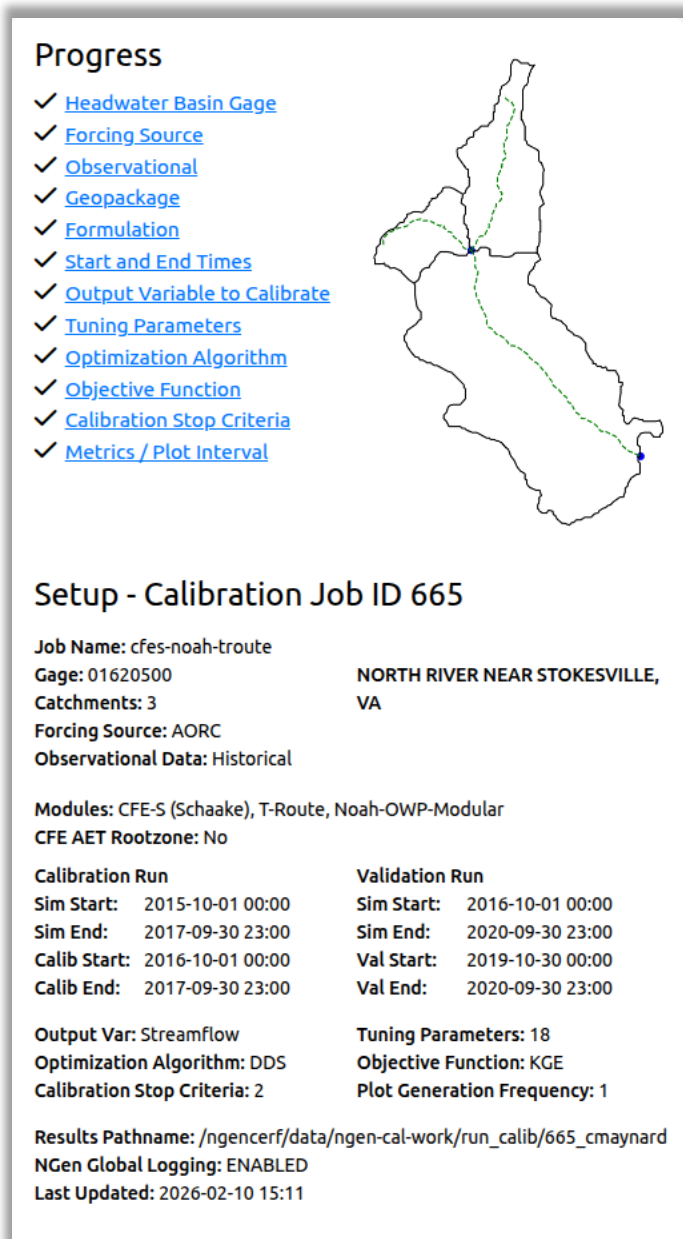
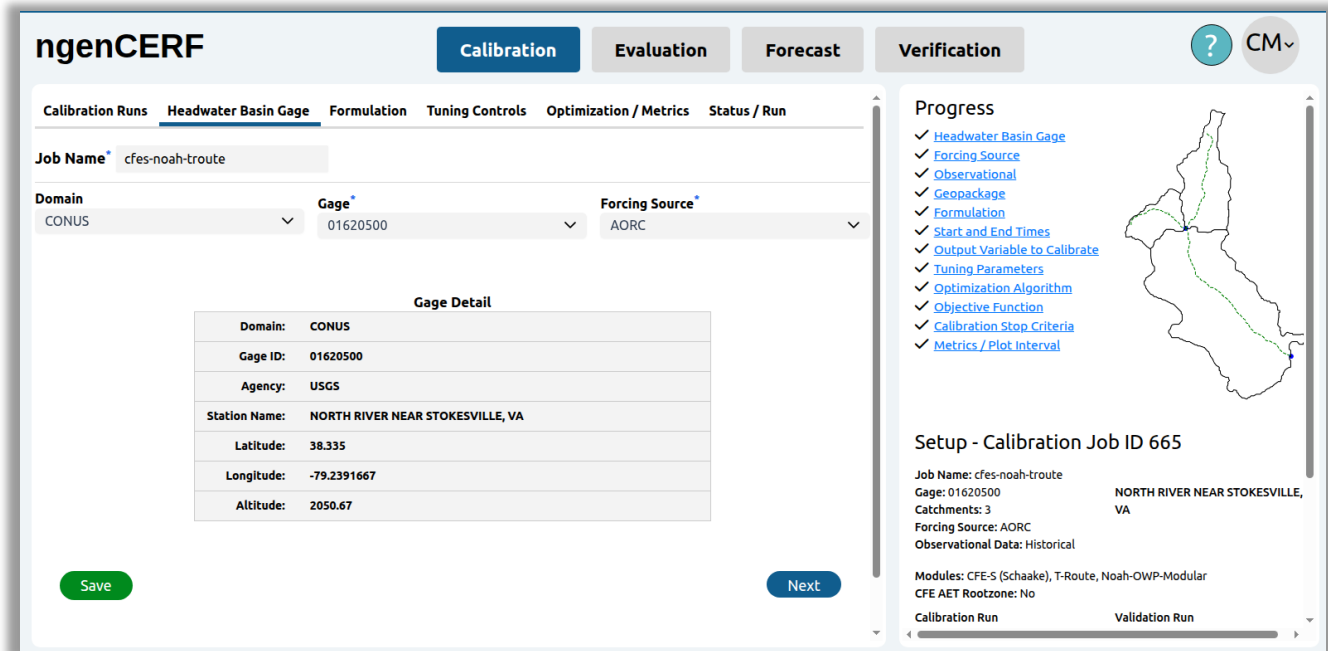


Figure 9 Calibration: Progress Area

5.4.3 Headwater Basin Gage Tab

The Headwater Basin Gage tab allows the user to select the headwater basin gage, forcing data, observed streamflow data (Historical only), and the GeoPackage (Hydrofabric only). The GeoPackage is automatically retrieved from **Data Services** based on the selected gage. Only gages identified by OWP as calibratable headwater basins are available for selection.



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Calibration Runs **Headwater Basin Gage** Formulation Tuning Controls Optimization / Metrics Status / Run

Job Name* cfes-noah-troute

Domain: CONUS Gage*: 01620500 Forcing Source*: AORC

Gage Detail	
Domain:	CONUS
Gage ID:	01620500
Agency:	USGS
Station Name:	NORTH RIVER NEAR STOKESVILLE, VA
Latitude:	38.335
Longitude:	-79.2391667
Altitude:	2050.67

Save Next

Progress

- ✓ Headwater Basin Gage
- ✓ Forcing Source
- ✓ Observational
- ✓ Geopackage
- ✓ Formulation
- ✓ Start and End Times
- ✓ Output Variable to Calibrate
- ✓ Tuning Parameters
- ✓ Optimization Algorithm
- ✓ Objective Function
- ✓ Calibration Stop Criteria
- ✓ Metrics / Plot Interval

Setup - Calibration Job ID 665

Job Name: cfes-noah-troute
 Gage: 01620500
 Catchments: 3
 Forcing Source: AORC
 Observational Data: Historical

Modules: CFE-S (Schaake), T-Route, Noah-OWP-Modular
 CFE AET Rootzone: No

Calibration Run Validation Run

Figure 10 Calibration → Headwater Basin Gage Tab

User Actions:

Job Name*

User's name for the job. Alphanumeric characters, underscore (_) and dashes (-) are allowed. Spaces are not allowed.

Domain

Select a domain to filter the list of gages in the Gage field. Options are CONUS, Alaska, Hawaii, and Puerto Rico/US Virgin Islands.

Gage*

Select or enter the gage ID for the Headwater Basin of interest.

Forcing

AORC (default) or NWM Retrospective.

Save

Save the selections made on this tab. Can be done periodically or after completing all selections.

Revert

This will restore the tab to the last saved selections. Appears when changes have been made to saved data.

Next

Move to the Formulation tab. If there is unsaved data, the user will be prompted to save before proceeding.

* An asterisk indicates a required field

When the Save button is clicked, the subsetted GeoPackage, forcing data, and observational data is retrieved from **Data Services**. The Observational data source is determined by the owner/agency of the gage. A basin image is generated, and the appropriate checkmarks in the **Progress** area are updated once the data has been successfully received and saved. This process may take several seconds.

If a gage does not appear in the list, it may not be an OWP NWM 3 Calibratable Headwater Basin Gage. For a complete list of over 1,600 calibratable headwater basin gages — including domain, NWS ID, NWM Reach ID, agency, and station name—see the [NGWPC Headwater Basin Gage Confluence Page](#).

5.4.4 Formulation Tab

The **Formulation** tab allows the user to define a named formulation for the calibration job and select the stack of hydrologic modules to use for calibration and validation. Modules are grouped by the process they represent. These are Glacier, Snowmelt, Evapotranspiration, Soil Moisture, Rainfall Runoff, and Routing. Selecting a module automatically highlights its associated group(s) and displays which NWM v3 output variables will be generated. The CFE Rootzone option is shown when a CFE module is selected.

A formulation is characterized as **Calibratable** if it meets the selection rules for each process, and **Complete** if it is calibratable and produces all required NWM v3 Output Variables. Formulations that are not calibratable **cannot be run**. The tab provides visual feedback on formulation completeness and calibratability as modules are selected. Users can save partial selections, revert changes, or navigate to other tabs using the **Prev** and **Next** buttons, by selecting tabs directly, or via links in the **Progress** area.

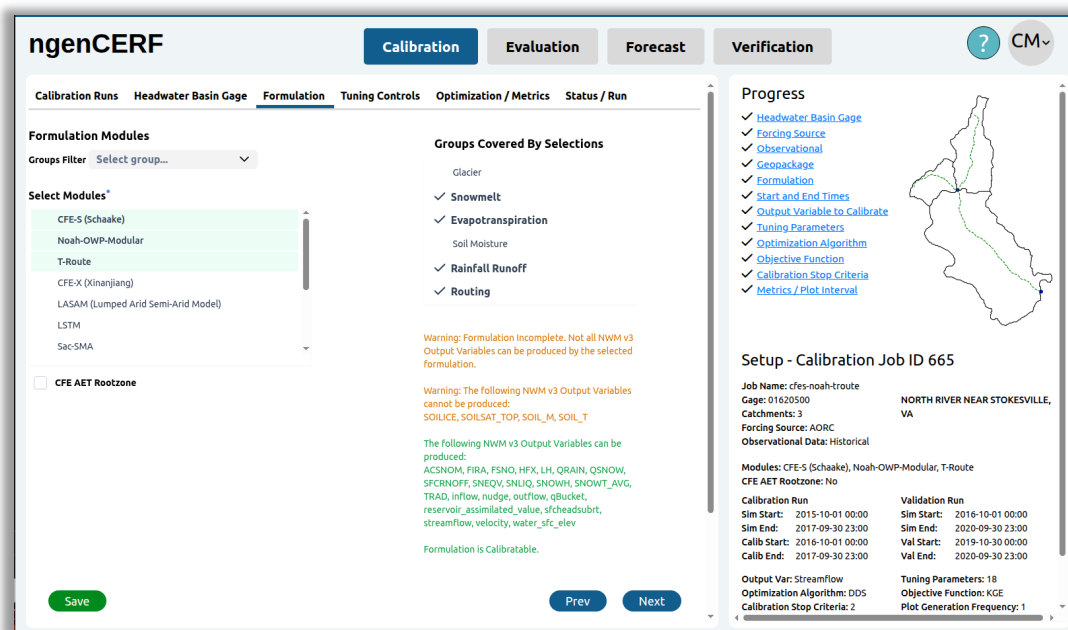


Figure 11 Calibration → Formulation Tab

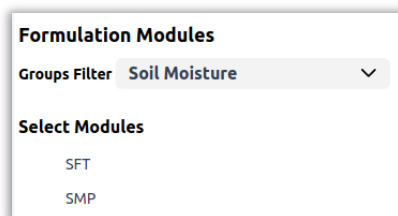


Figure 12 Calibration → Formulation Tab Group Filter

User Actions:

Group	Filter the module list by the group classification.
Select Modules*	User selects the modules for the formulation.
Save	Click this button to save the data entered on the tab. This can be done periodically or after making all the selections.
Revert	Click this button to revert selections for this tabbed work area to the saved data. It appears if a saved selection changes.
Prev	Move to the Headwater Basin Tab. If there is any unsaved data, the user will first be prompted to save.
Next	Move to the Tuning Controls Tab. If there is any unsaved data, the user will first be prompted to save.

* An asterisk indicates a required field

5.4.4.1 Formulation Rules:

See also [Formulation Rules Confluence Page](#)

- Only one module per process (except for SFT and SMP, which are always run together currently)
- Processes are Evapotranspiration, Snowmelt, Rainfall Runoff, Auxiliary Soil (aka Soil Moisture), and Routing

Evapotranspiration Processes (Required, except for LSTM)

- noah-owp-modular:
 - If noah-owp-modular is selected for evapotranspiration, it is also selected as the snow model if another single purpose snow model has not been selected.
 - If noah-owp-modular has been selected as the snow model, and another single purpose evapotranspiration module has been selected, warn user that there are two evapotranspiration modules selected and noah-owp-modular will not be used for evapotranspiration.
- potential-evapotranspiration (PET): evapotranspiration
 - Not currently integrated in ngenCERF

Snowmelt Processes (Optional)

- Utah Energy Balance (UEB):
 - If selected, will override snowmelt component of noah-owp-modular
 - Cannot be selected in conjunction with another single purpose snow model
- SNOW-17
- noah-owp-modular:
 - If noah-owp-modular is selected for snowmelt, it is also selected as the evapotranspiration module if another single purpose evapotranspiration module has not been selected.
 - If noah-owp-modular has been selected as the evapotranspiration module, and another single purpose snow-melt module has been selected, user is

warned that there are two snow-melt modules selected and noah-owp-modular will not be used for snowmelt.

Rainfall Runoff Processes (Required)

- CFE-S, CFE-X
- SAC-SMA
- Topmodel
- LASAM
- LSTM
 - If LSTM is selected as the rainfall runoff model the snow-melt, evapotranspiration, and auxiliary soil processes will be disabled.

Auxiliary Soil Processes (Optional)

Aka Soil Moisture Group

Generally required to produce all NWM output variables.

- SoilFreezeThaw (SFT)
- SoilMoistureProfiles(SMP)
 - SFT is paired with SMP (required).
 - They require a rainfall-runoff model for input (CFE-X/S, LASAM, and TopModel currently have integrations).
 - They share parameters with CFE and LASAM, so when paired with either of those modules the calibration of parameters is done jointly..

Routing (Required)

- T-Route

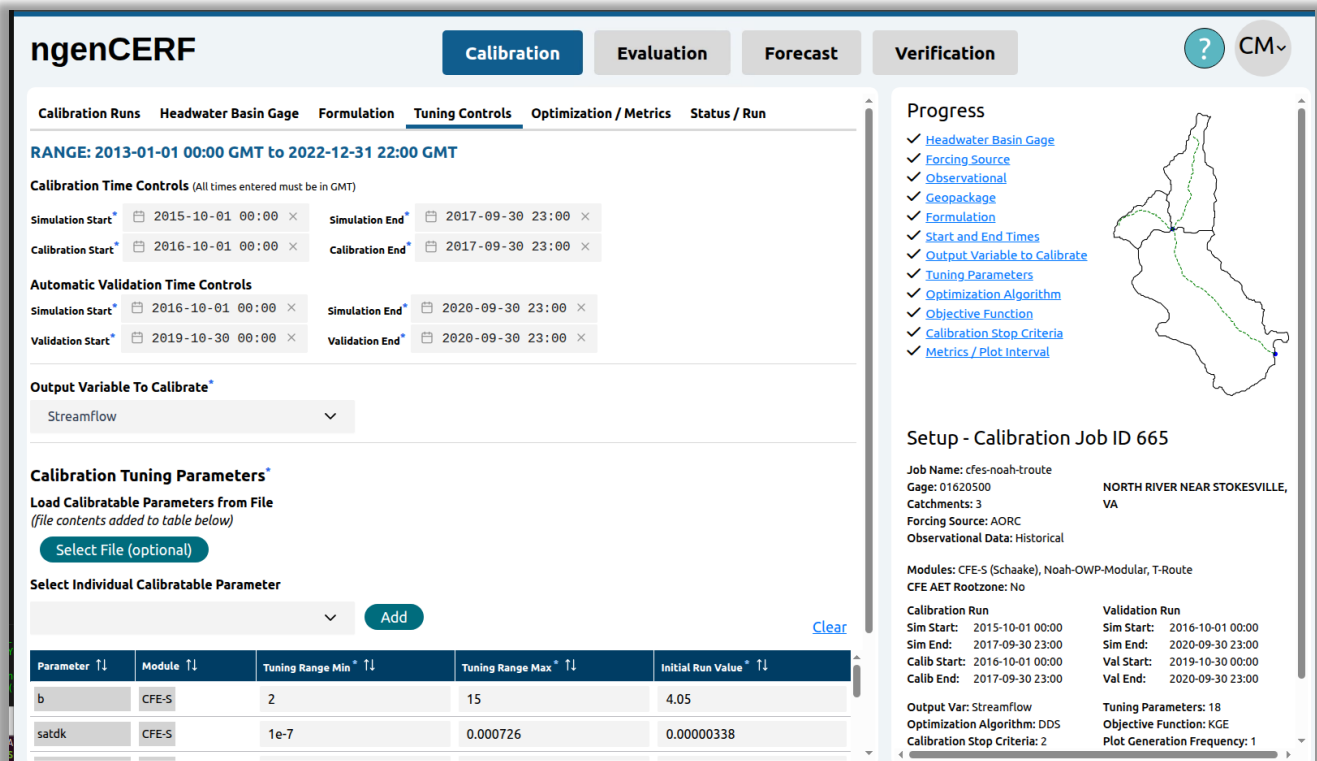
5.4.5 Tuning Controls Tab

The **Tuning Controls** tab allows the user to define the time periods for calibration and automatic validation, select the output variable to calibrate, and specify which formulation parameters to tune during calibration. When LSTM is included in the formulation, only the time controls are required.

Calibration and validation times controls **each require four times**: the **Simulation Period Start and End** and the **Calibration or Validation Period Start and End**. The simulation start times provide a “warm-up” period to initialize model states before official calibration or validation begins. All time selections are constrained by the intersection of available forcing and observational data, must be hourly, and the validation period cannot overlap the calibration period.

The Output Variable to Calibrate dropdown and Tuning Parameters are **enabled** once a formulation is selected, with streamflow currently being the only option available for the output variable. Calibration Tuning Parameters are based on the selected formulation and can be added from the dropdown or uploaded from a file. Parameters in the table require a minimum, maximum, and initial value, with the initial value used for iteration 0 of calibration, also called the “control run”. Warnings are provided if values fall outside allowed ranges, but the user may proceed.

Users can save partial selections, revert changes, or navigate to other tabs using the Prev and Next buttons or links in the Progress area. Data must be entered into the field using Enter or, in the case of the date picker, Select, otherwise values will be discarded. Time controls and parameters are disabled until the required data and formulation selections have been made.



ngenCERF Calibration Evaluation Forecast Verification

Calibration Runs Headwater Basin Gage Formulation **Tuning Controls** Optimization / Metrics Status / Run

RANGE: 2013-01-01 00:00 GMT to 2022-12-31 22:00 GMT

Calibration Time Controls (All times entered must be in GMT)

Simulation Start* 2015-10-01 00:00 x Simulation End* 2017-09-30 23:00 x
 Calibration Start* 2016-10-01 00:00 x Calibration End* 2017-09-30 23:00 x

Automatic Validation Time Controls

Simulation Start* 2016-10-01 00:00 x Simulation End* 2020-09-30 23:00 x
 Validation Start* 2019-10-30 00:00 x Validation End* 2020-09-30 23:00 x

Output Variable To Calibrate*

Streamflow

Calibration Tuning Parameters*

Load Calibratable Parameters from File
 (file contents added to table below)

Select File (optional)

Select Individual Calibratable Parameter

Parameter	Module	Tuning Range Min	Tuning Range Max	Initial Run Value
b	CFE-S	2	15	4.05
satdk	CFE-S	1e-7	0.000726	0.00000338

Progress

- ✓ Headwater Basin Gage
- ✓ Forcing Source
- ✓ Observational
- ✓ Geopackage
- ✓ Formulation
- ✓ Start and End Times
- ✓ Output Variable to Calibrate
- ✓ Tuning Parameters
- ✓ Optimization Algorithm
- ✓ Objective Function
- ✓ Calibration Stop Criteria
- ✓ Metrics / Plot Interval

Setup - Calibration Job ID 665

Job Name: cfe-s-noah-trout
 Gage: 01620500 NORTH RIVER NEAR STOKESVILLE, VA
 Catchments: 3
 Forcing Source: AORC
 Observational Data: Historical

Modules: CFE-S (Schaake), Noah-OVP-Modular, T-Route
 CFE AET Rootzone: No

Calibration Run
 Sim Start: 2015-10-01 00:00
 Sim End: 2017-09-30 23:00
 Calib Start: 2016-10-01 00:00
 Calib End: 2017-09-30 23:00

Validation Run
 Sim Start: 2016-10-01 00:00
 Sim End: 2020-09-30 23:00
 Val Start: 2019-10-30 00:00
 Val End: 2020-09-30 23:00

Output Var: Streamflow
 Optimization Algorithm: DDS
 Calibration Stop Criteria: 2

Tuning Parameters: 18
 Objective Function: KGE
 Plot Generation Frequency: 1

Figure 13 Calibration → Tuning Controls Tab

General Time Control Rules

- Calibration Time Controls
 - Calibration period must fall within the Calibration Simulation period.
 - Calibration period cannot overlap the Validation period.
- Validation Time Controls
 - Validation Simulation period must encompass the entire Calibration Simulation period.
 - Validation period must fall within the Validation Simulation period.
- Hourly requirement
 - All times are in full hours and must be specified at the top of the hour (hh:00).

User Actions:

Calibration Time Controls

Calibration Simulation Period

Simulation Start*

Date and hour to start the calibration simulation

Simulation End*

Date and hour to stop the calibration simulation

Calibration Period Calibration Start to Calibration End - **Cannot overlap the Validation period.**

Calibration Start Date and hour to start calibrating.

Calibration End Date and hour to stop calibration.

Automatic Validation Time Controls

Validation Simulation Period Simulation Start to Simulation End - **Must encompass the entire Calibration period.**

Simulation Start* Date and hour to start the validation simulation - **Must be outside of the Calibration Start and End period.**

Simulation End* Date and hour to stop the validation simulation

Validation Period Validation Start to Validation End - **Cannot overlap the Calibration period.**

Validation Start* Date and hour to start validation.

Validation End* Date and hour to stop validation.

Output Variable to Calibrate³ Select an output variable from the pulldown menu. Currently Streamflow is the only support output for calibration.

Calibration Tuning Parameters Area³

At least on tuning parameters is required.

Select File **Load Calibratable Parameters From File** Optionally upload a file of preselected calibratable parameters associated with the formulation. This file can be either CSV, tab or space delimited and must contain the parameter name, module name, minimum value, maximum value, and initial value. See Figure 14 for an example CSV file.

This file is read and its contents added to the Parameter table where the parameters can be modified or deleted.

Add **Select Individual Calibratable Parameter.** Button to add the selected calibratable parameter from the pulldown menu. This menu is populated with calibratable parameters based on the modules in the formulation.

Parameter Table Editable list of the calibration parameters

- Parameters may be added from the Calibratable Parameters dropdown or loaded from an uploaded file.
- Parameter and Module are un-editable fields.
- Tuning Range Min, Tuning Range Max, and Initial Run Value are required, editable fields.
- The Initial Run Value is used for the calibration control run (iteration 0) and may fall outside the specified minimum/maximum range.

- If the Initial Run Value is outside this range, it will be saved and a warning will be issued; the user may either proceed or modify the value and save again.
- **To remove a parameter, right-click** it in the table and select Delete.

Save	Select this button to save the data entered on the tab. This can be done periodically or after making all the selections.
Revert	Select this button to reset selections for this tabbed work area.
Prev	Move to the Formulation Tab. If there is any unsaved data, the user will first be prompted to save.
Next	Move to the Optimization / Metrics Tab. If there is any unsaved data, the user will first be prompted to save

* An asterisk indicates a required field

See the [BMI Exchange Items and Module Parameters Confluence Page](#) for a complete list of Calibratable Parameters per module and Section 6.4 for more details.

```
param,min,max,init,model
field_capacity,10.5,500.0,20.0,LASAM
hydraulic_conductivity,1.0,50.0,10.0,LASAM
smcmax,0.4,0.7,0.439,LASAM
```

Figure 14 Calibration: Example Calibratable Parameters CSV file

Important:

- Tuning Controls cannot be selected until both the Headwater Basin Gage tab data and the Formulation are selected. If the Headwater Basin Gage is changed, the times entered for the Tuning controls will be cleared. If the Formulation is changed, the Calibration Tuning Parameters will be updated, removing any parameters for modules that were removed from the Formulation
- If the headwater basin gage, forcing data or observational data is changed, the times entered for the Calibration and Validation will be cleared.
- If the formulation of modules selected is changed, the Calibration Tuning Parameters will be cleared.

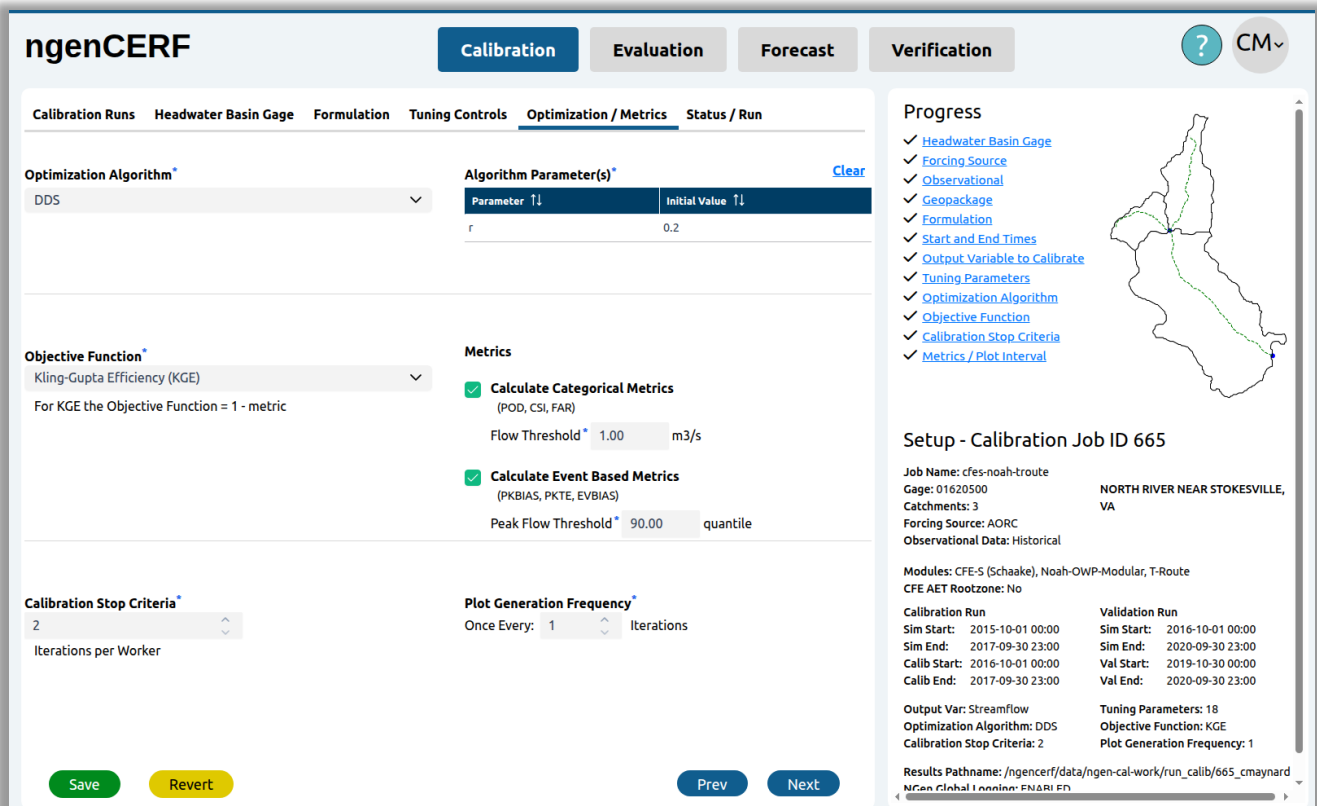
5.4.6 Optimization/Metrics Tab

The **Optimization / Metrics** tab allows the user to configure how the calibration manager runs the calibration. This includes selecting the, specifying Algorithm Parameters, choosing the Objective Function, deciding whether to calculate Categorical Metrics or Event-Based Metrics, setting the Calibration Stop Criteria, and defining the Plot Generation Frequency.

By default, categorical metrics (POD, CSI, FAR) and event-based metrics (PKBIAS, PKTE, EVBIAS) are not calculated, but users can opt to compute them. Whenever a categorical or

event-based metric is selected, either as the objective function or via its checkbox, the corresponding threshold must be provided: a Flow Threshold for categorical metrics and a Peak Flow Threshold for event-based metrics.

This tab is not available when LSTM is included in the formulation. Users can navigate between tabs using the Prev and Next buttons or select tabs directly from the Progress area. Partial saves are allowed at any time using the Save button.



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Calibration Runs Headwater Basin Gage Formulation Tuning Controls **Optimization / Metrics** Status / Run

Optimization Algorithm*
DDS

Algorithm Parameter(s)* [Clear](#)

Parameter	Initial Value
r	0.2

Objective Function*
Kling-Gupta Efficiency (KGE)
For KGE the Objective Function = 1 - metric

Metrics

- ☒ Calculate Categorical Metrics (POD, CSI, FAR)
Flow Threshold* 1.00 m3/s
- ☒ Calculate Event Based Metrics (PKBIAS, PKTE, EVBIAS)
Peak Flow Threshold* 90.00 quantile

Calibration Stop Criteria*
2
Iterations per Worker

Plot Generation Frequency*
Once Every: 1 Iterations

[Save](#) [Revert](#) [Prev](#) [Next](#)

Progress

- ✓ Headwater Basin Gage
- ✓ Forcing Source
- ✓ Observational
- ✓ Geopackage
- ✓ Formulation
- ✓ Start and End Times
- ✓ Output Variable to Calibrate
- ✓ Tuning Parameters
- ✓ Optimization Algorithm
- ✓ Objective Function
- ✓ Calibration Stop Criteria
- ✓ Metrics / Plot Interval

Setup - Calibration Job ID 665

Job Name: cfes-noah-trout
Gage: 01620500
Catchments: 3
Forcing Source: AORC
Observational Data: Historical

Modules: CFE-S (Schaake), Noah-OVP-Modular, T-Route
CFE AET Rootzone: No

Calibration Run	Validation Run
Sim Start: 2015-10-01 00:00	Sim Start: 2016-10-01 00:00
Sim End: 2017-09-30 23:00	Sim End: 2020-09-30 23:00
Calib Start: 2016-10-01 00:00	Val Start: 2019-10-30 00:00
Calib End: 2017-09-30 23:00	Val End: 2020-09-30 23:00

Output Var: Streamflow
Optimization Algorithm: DDS
Calibration Stop Criteria: 2

Tuning Parameters: 18
Objective Function: KGE
Plot Generation Frequency: 1

Results Pathname: /ngenCERF/data/ngen-cal-work/run_calib/665_cmaynard
NGen Global License: ENRI EN

Figure 15 Calibration → Optimization Metrics Tab

User Actions:

Optimization Algorithm*

Select the optimization algorithm from the pulldown. Options are DDS, GWO, and PSO. DDS has one worker while GWO and PSO have multiple workers.

Algorithm Parameters*

Based on the Optimization Algorithm chosen, this table will be auto populated, and the user may update the default values.

Objective Function*

User selects a metric from the list of metrics in the pulldown. If the metric chosen is a Categorical Metric (POD, CSI, FAR) then the user must supply the Flow Threshold Value in m³/s units.

If the metric chosen is an Event Based Metric (PKBIAS, PKTE, EVBIAS) then the user must supply the Peak Flow Threshold Value in quartile units.

Flow Threshold	Required input in m ³ /s units if a categorical metric is selected as the Objective Function.
Peak Flow Threshold	Required input in quartile units if an event-based metric is selected as the Objective Function.
Additional Metrics	Select additional metrics to be calculated. All other metrics are calculated by the calibration manager except Categorical and Event Based metrics.
Calculate Categorical Metrics	Checkbox to optionally calculate the categorical metrics POD, CSI and FAR. If checked the user must supply the dependent input. The user must supply the Flow Threshold.
Flow Threshold	Required input in m ³ /s units if Calculate Categorical Metrics checkbox selected.
Calculate Event Based Metrics	Checkbox to optionally calculate the event-based metrics PKBIAS, PKTE, EVBIAS. The user must supply the Peak Flow Threshold.
Peak Flow Threshold	Required input in quartile units if Calculate Event Based Metrics checkbox is selected.
Calibration Stop Criteria*	Select the number of Worker Iterations to execute before stopping the calibration.
Plot Generation Frequency*	Number of iterations to run before generating a new set of plots that can be viewed by the user during the calibration run. These plots are cumulative and include data from all previous iterations.
Save	Select this button to save the data entered on the tab. This can be done periodically or after making all selections.
Revert	Select this button to reset selections for this tabbed work area.
Prev	Move to the Tuning Controls Tab. If there is any unsaved data, the user will first be prompted to save.
Next	Move to the Run / Status Tab. If there is any unsaved data, the user will first be prompted to save

* An asterisk indicates a required field

Table 6: Calibration Metrics

Metric Acronym	Description
Corr	Pearson Correlation
CSI	Critical Success Index (Categorical Metric)
EVBIAS	Event Volume Bias (Event Based Metric)
FAR	False Alarm Ratio (Categorical Metric)

Metric Acronym	Description
HSEG_FDC	Percent bias of high flow segment of flow duration curve
KGE	Kling-Gupta Efficiency
LSEG_FDC	Percent bias of low flow segment of flow duration curve
MAE	Mean Absolute Error
NNSE	Normalized NSE
NSE	Nash-Sutcliffe-Efficiency
NSELog	NSE of logarithmic values
PBIAS	Percent Bias
PKBIAS	Event Absolute Peak Flow Bias (Event Based Metric)
PKTE	Event Peak Flow Timing Error (Event Based Metric)
POD	Probability of Detection (Categorical Metric)
RMSE	Root Mean Square Error
RSR	Ratio of RMSE to standard deviation of observation

5.4.7 Run / Status Tab

The **Run / Status** tab allows users to review their calibration setup, start a calibration run, and monitor progress through calibration and validation. This tab supports both active job execution and post-run review of results, plots, and logs.

The right side of the screen summarizes progress through the calibration workflow and reflects the current job status. Once all required setup steps are complete, the job status becomes Ready, enabling the Run button.

When Run is selected, the system prepares and validates the required data, submits the job for execution, and begins the calibration process when resources are available. During this time, users can track job status, elapsed time, and completed iterations in real time.

As the job runs:

- **Logs** become available as soon as execution begins and can be viewed throughout the run.
- **Iteration progress** is updated as each iteration completes. Iteration 0 represents the control run.
- **Calibration plots** become available after the first iteration and update automatically based on the selected plot interval.
- **Validation plots** are available once validation runs complete.

After calibration and validation finish successfully, the job status changes to Done, and users may continue reviewing plots and logs or navigate to the Evaluation workflow for additional analysis, including performance metrics and time series data. Jobs may be canceled at any time while running.

If a job fails at any stage, relevant log files are automatically displayed to assist with troubleshooting, and all available logs remain accessible from the display menu.

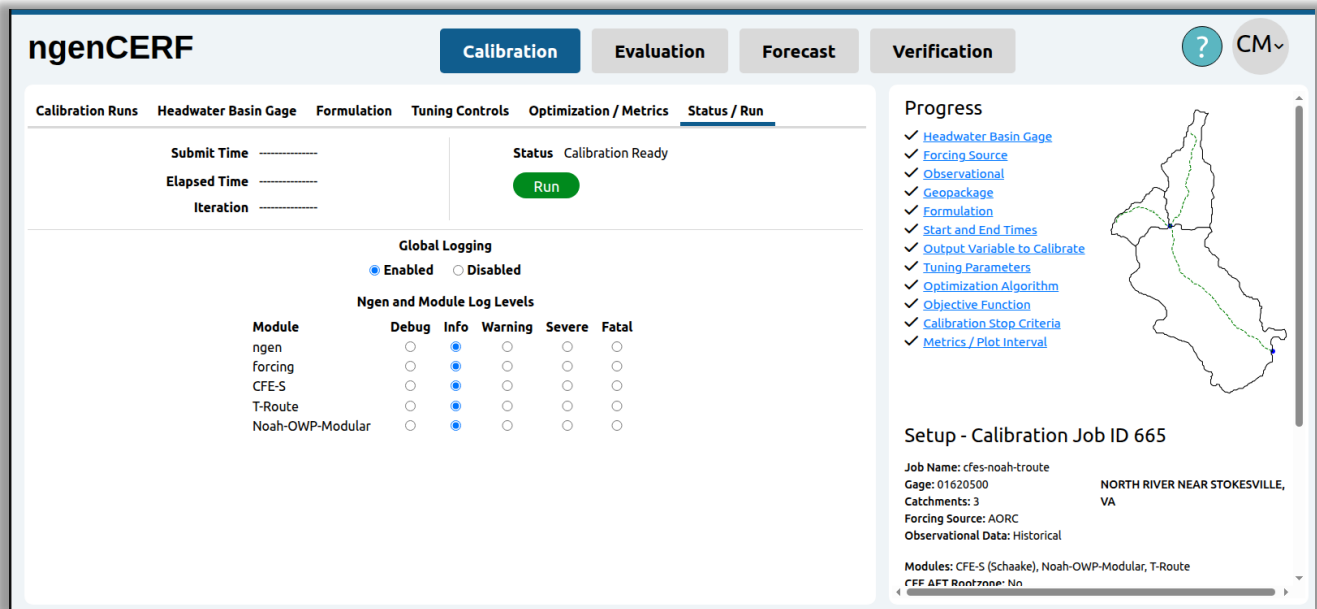


Figure 16 Calibration → Status/Run Tab View

User Actions:

Run

If the status is Calibration Ready, clicking this button starts the calibration run. If it is not ready, something is missing from the calibration setup.

Global Logging

Allows the user to Enable (default) or Disable logging in ngen and the formulation modules.

Log Levels

Select the log level for ngen and each module in the formulation. Default log level is INFO.

Running Tab Views

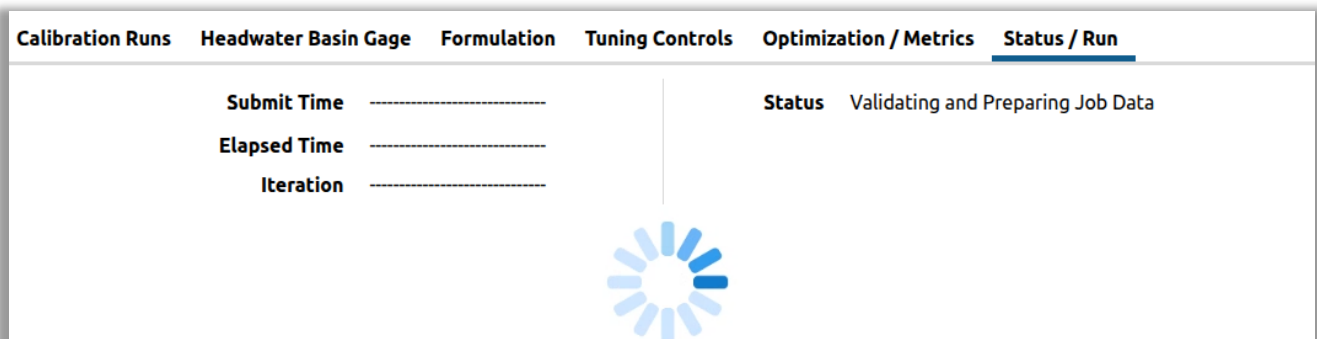


Figure 17 Calibration → Status/Run Tab View: Validating and Preparing Job Data

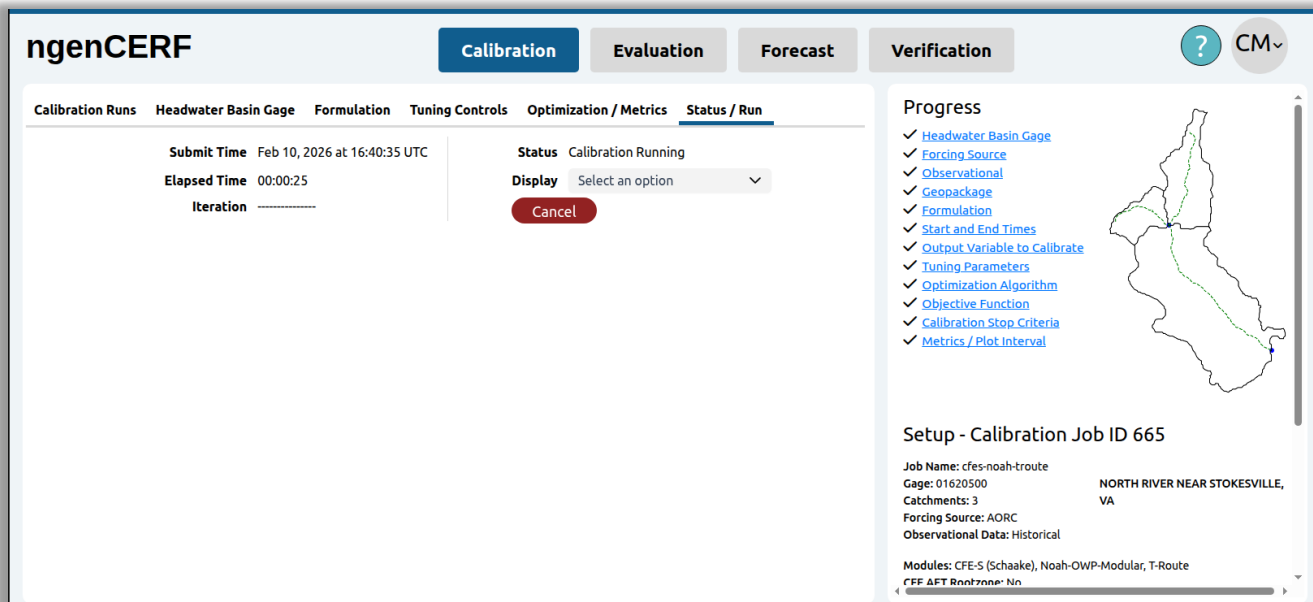


Figure 18 Calibration → Status/Run Tab View: Running Before Plots or Logs Available

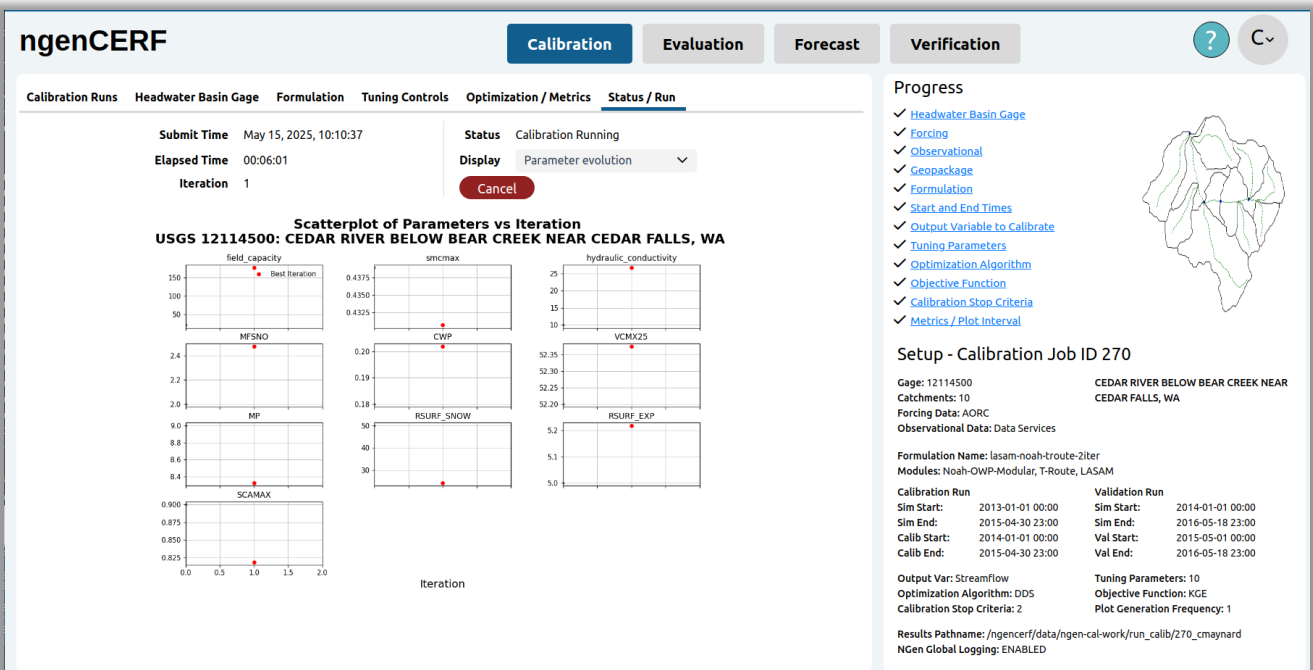


Figure 19 Calibration → Status/Run Tab View: After Iteration 1 Complete

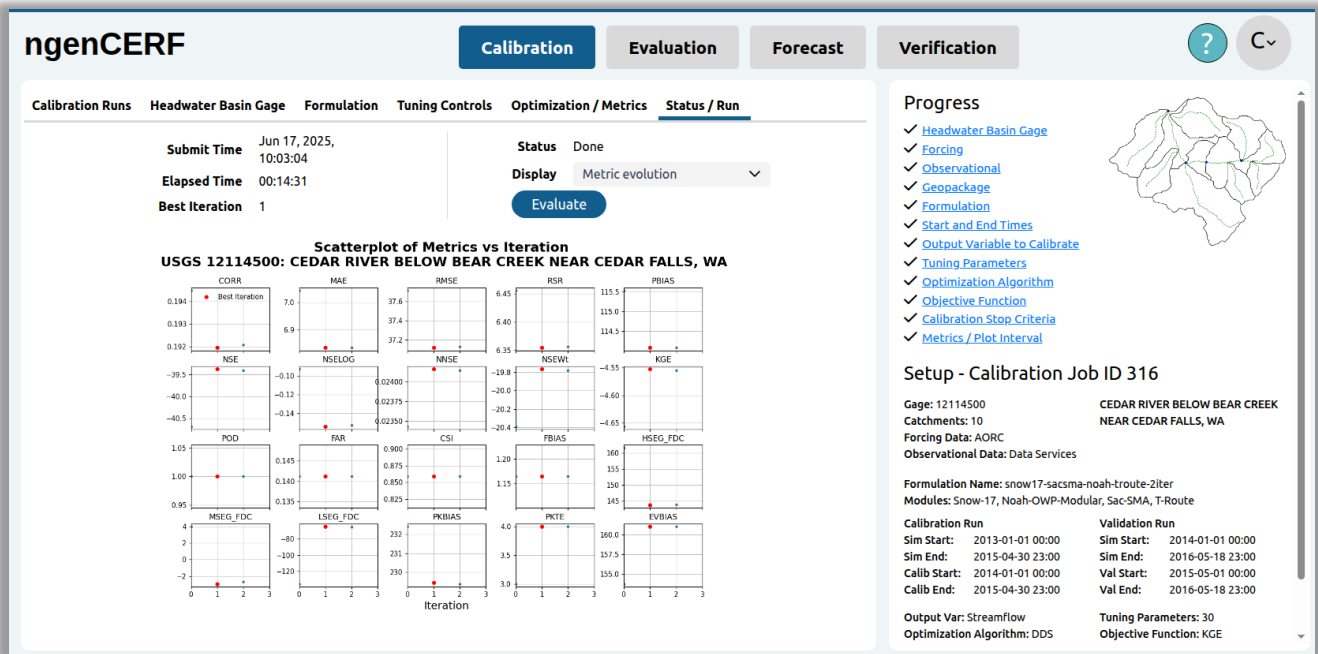


Figure 20 Calibration → Status/Run Tab View: Job Complete

User Actions:

Display

Select a plot or log to display from the pulldown menu. These plots are updated at the Plot Generation Frequency and are cumulative. Logs are available once the job starts running. Calibration plots are available after iteration 0. Validation plots are available when the job is done.

Cancel

Cancel the currently running calibration. Depending on when this button is clicked, the Automatic Validation Run will not be executed.

Evaluate

When the job successfully completes, this button appears and takes the user to the Evaluation workflow where further evaluation and permutation comparison can be done.

5.5 Evaluation Workflow

The Evaluation interface allows users to review results from completed calibration and validation runs. The calibration manager retains data generated for the **control run**, the **final iteration**, and the **best parameter set**, which are available for immediate review.

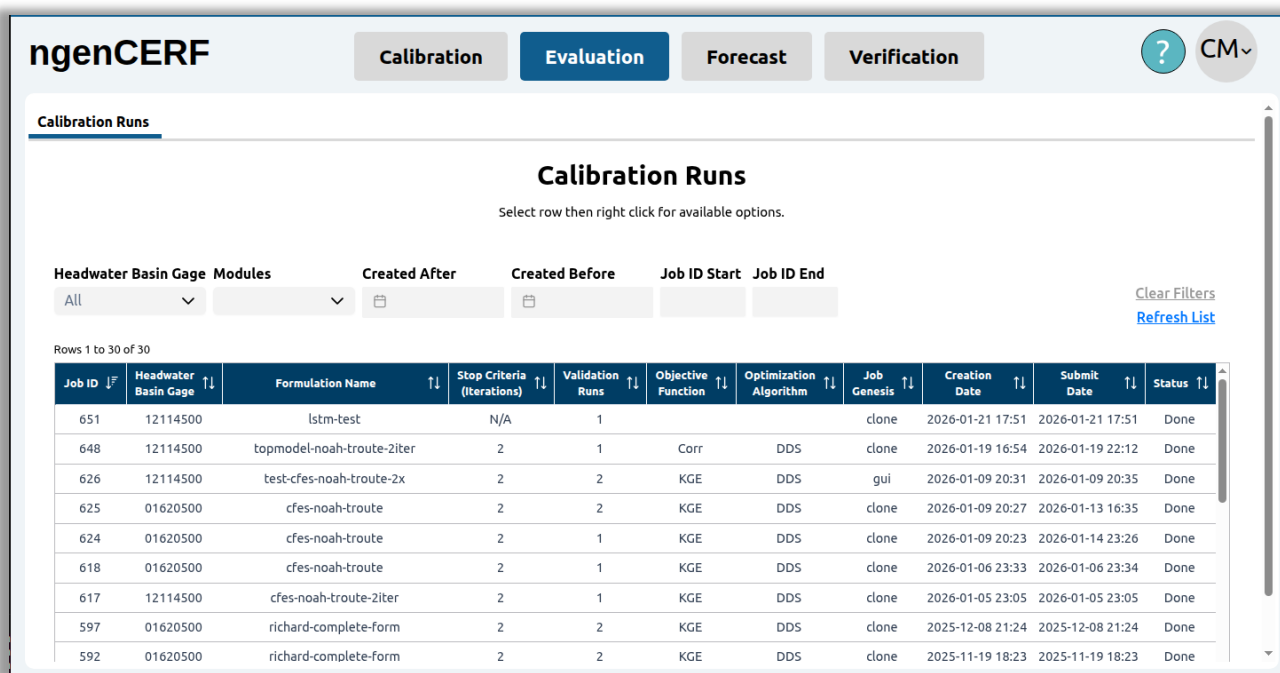
After selecting a calibration run, users can examine plots, metrics, and other outputs in the Evaluate tab. If results from a different calibration iteration are needed, an additional validation can be executed by selecting a parameter set in the **Select Alternate Iteration** tab. This process regenerates the required output data for evaluation.

5.5.1 Calibration Runs Tab

The **Calibration Runs** tab in the Evaluation workflow displays a table of successfully completed calibration jobs available for review and analysis. Users can filter, sort, and select calibration jobs to quickly locate runs of interest.

Jobs can be selected individually or in groups using standard keyboard controls. Context-sensitive actions are available through a right-click menu, allowing users to evaluate results, compare permutations (formulations) for the same gage, initiate additional validation runs for alternate iterations, download results, or view calibration details. When a calibration job includes multiple validation runs, users can select which validation run to evaluate.

The tab provides flexible filtering options based on gage, modules, creation date, and job ID range, enabling efficient navigation of large job lists. Archived jobs are hidden from the table and cannot be evaluated until they are unarchived from the Calibration workflow.



Job ID	Headwater Basin Gage	Formulation Name	Stop Criteria (Iterations)	Validation Runs	Objective Function	Optimization Algorithm	Job Genesis	Creation Date	Submit Date	Status
651	12114500	lstm-test	N/A	1			clone	2026-01-21 17:51	2026-01-21 17:51	Done
648	12114500	topmodel-noah-troute-2iter	2	1	Corr	DDS	clone	2026-01-19 16:54	2026-01-19 22:12	Done
626	12114500	test-cfes-noah-troute-2x	2	2	KGE	DDS	gui	2026-01-09 20:31	2026-01-09 20:35	Done
625	01620500	cfes-noah-troute	2	2	KGE	DDS	clone	2026-01-09 20:27	2026-01-13 16:35	Done
624	01620500	cfes-noah-troute	2	1	KGE	DDS	clone	2026-01-09 20:23	2026-01-14 23:26	Done
618	01620500	cfes-noah-troute	2	1	KGE	DDS	clone	2026-01-06 23:33	2026-01-06 23:34	Done
617	12114500	cfes-noah-troute-2iter	2	1	KGE	DDS	clone	2026-01-05 23:05	2026-01-05 23:05	Done
597	01620500	richard-complete-form	2	2	KGE	DDS	clone	2025-12-08 21:24	2025-12-08 21:24	Done
592	01620500	richard-complete-form	2	2	KGE	DDS	clone	2025-11-19 18:23	2025-11-19 18:23	Done

Figure 21 Evaluation → Calibration Runs Tab

User Actions: Filters

Filters include Headwater Basin Gage, Modules, creation date range, and Job ID range. When multiple filters are applied, only jobs that satisfy all active filters are shown.

The Modules filter supports multi-selection. When more than one module is selected, the Module Match option allows the user to specify whether jobs must include all selected modules or any of the selected modules.

The table updates automatically as filter selections change.

Filters can be cleared at any time using Clear Filters, and Refresh List reloads the current filtered results.

Context Menus:

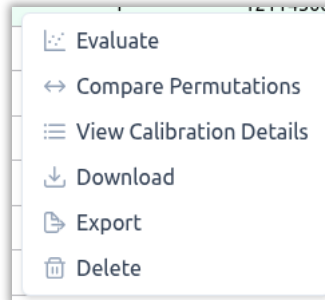


Figure 22 Evaluation → Calibration Job Context Menu: 1 Validation Run

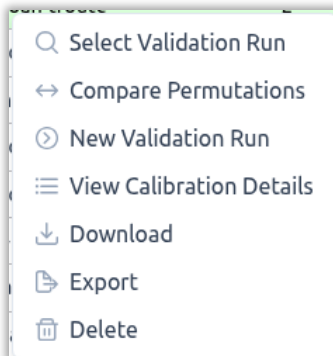


Figure 23 Evaluation → Calibration Job Context Menu: Multiple Validation Run

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Calibration

Evaluation

Forecast

Verification

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Calibration Runs

Select Alternate Iteration

Validation Runs for Calibration Job 242

Select row then right click for available options.

Validation Job ID	Iteration	Status	Submit Date	b	satdk	satpsi	slope	max_gw_storage	Cgw	expon	Klf	RSURF_EXP	MFSNO	CWP	VCMX25	MP	RSURF_SNOW	SCAMAX
255	1	Done	2025-05-15 17:39	8.40525	0.00011	0.53245	0.23698	0.13302	0.00001	5.69076	0.01822	5.27263	2.44349	0.09872	48.79171	9.83378	39.38996	0.92173
232	44*	Done	2025-05-08 01:48	3.43006	0.00005	0.47000	0.14147	0.14721	0.00120	4.56831	0.09275	5.63487	2.71419	0.12073	65.97048	9.36648	47.85992	0.83939

* Iteration containing best parameter set determined during calibration.

Return to Calibration Jobs

Figure 24 Evaluation → Calibration Runs, Calibration Job, Select Validation Run

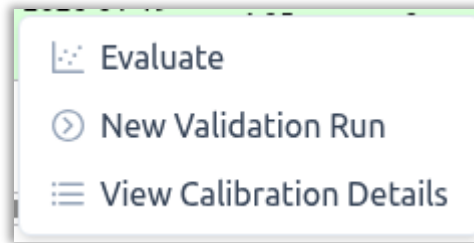


Figure 25 Evaluation → Calibration Job, Select Validation Run Context Menu

ngenCERF Calibration **Evaluation** Forecast Verification ? Cv

Calibration Runs Compare Permutations

Select Calibration Runs to Compare Compare

Headwater Basin Gage
12114500

Job ID	Status	Submit Date	Formulation Name	Headwater Basin Gage	Calibration Period	Objective Function	Optimization Algorithm
270	Done	2025-05-15 17:10	lasam-noah-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
268	Done	2025-05-14 19:19	lasam-noah-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
262	Done	2025-05-14 14:17	topmodel-noah-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
258	Done	2025-05-13 02:38	snow17-sacsma-noah-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
246	Done	2025-05-08 03:23	snow17-sacsma-noah-troute-100liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
242	Done	2025-05-07 23:56	cfes-noah-troute-50liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
238	Done	2025-05-06 21:09	cfes-noah-troute-10liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
214	Done	2025-05-01 23:13	topmodel-noah-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
211	Done	2025-04-25 04:55	ueb-noah-cfes-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
191	Done	2025-04-18 23:04	ueb_noah_sacsma_2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
188	Done	2025-04-30 23:41	lasam-noah-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS

Return to All Calibration Jobs

Figure 26 Evaluation → Calibration Runs, Select Jobs to Compare Permutations

Right Click Actions

Evaluate

Select from the following options:

Takes user to the Evaluate tab. Appears if calibration job has 0 or 1 validation jobs.

Select Validation Run

When there is more than one (1) validation job, this will replace the Evaluate option. Selecting this will take the user to the view showing the table of associated validation jobs.

Compare Permutations

Selecting this will take the user to the view showing the table of validation jobs with the same gage ID.

New Validation Run

After a calibration run has been selected this button will appear. Selecting this button will take the user to the Select Alternate Iteration to select the parameter set to use for another validation run.

View Calibration Details

View the summary details of the calibration job.

View Validation Run Status	View the run status of a validation run.
Download Calibration Data	Download all data associated with a Done calibration job.
Export	Export the Calibration Definition File. (Used by the CLI.)
Delete Calibration Job	Permanently deletes the selected calibration job(s) from the system.
Return To All Calibration Jobs	When on the view to select a Validation Run or select Calibration Jobs to compare, click this button to return to the table of Calibration Runs. Clicking the Calibrations Runs tab name will not work in these views.

5.5.2 Evaluate Tab

The user is provided with a dropdown to select plots generated during calibration and validation plots generated during the calibration as well as Time Series Viewer.

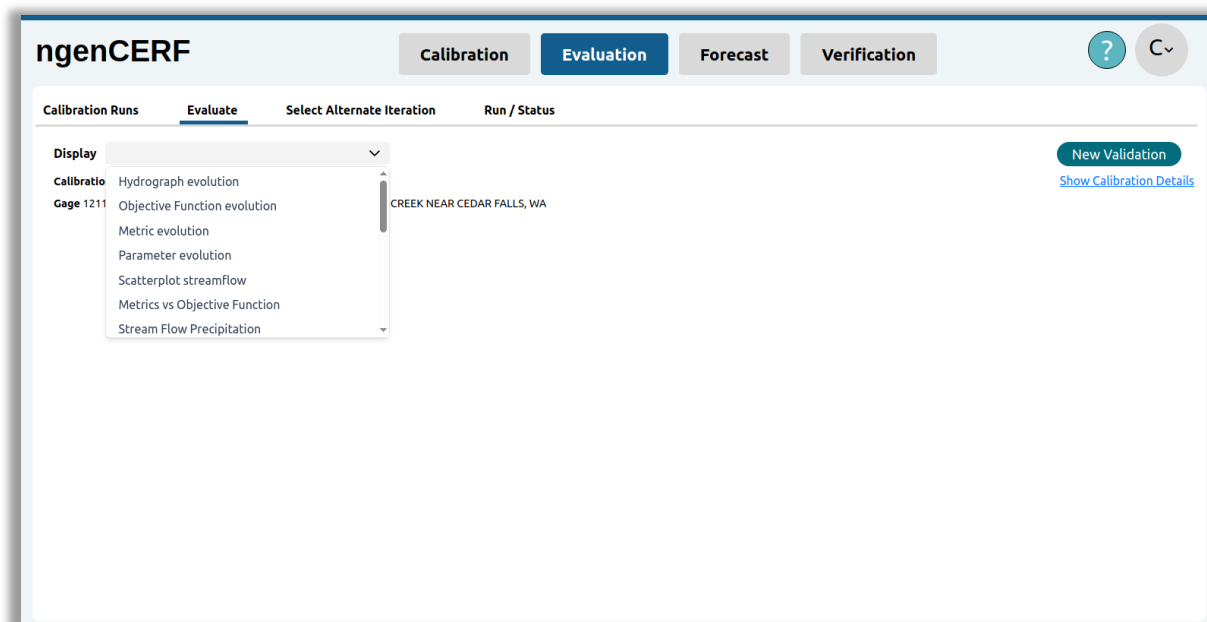


Figure 27 Evaluation → Evaluate Tab

5.5.2.1 Plots and Tables

The calibration manager program creates plot and saves iteration data. These are available for viewing side-by-side or in table form. For example, **Metric Evolution**.

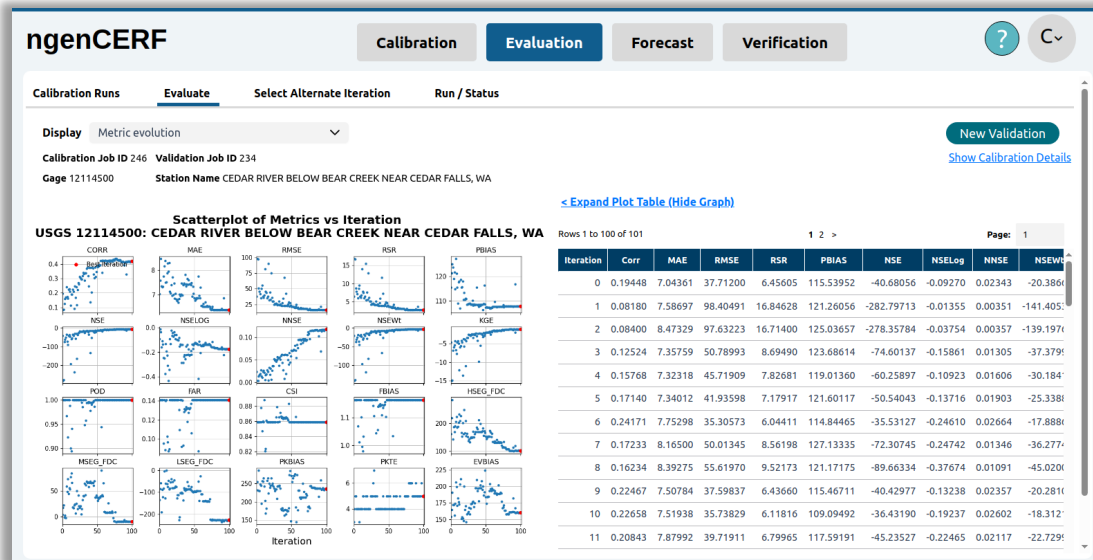


Figure 28 Evaluation → Evaluate Tab View: Plot and Table

5.5.2.2 Time Series Viewer

A time series viewer is available for plots with saved time series data, such as Hydrograph Evolution.

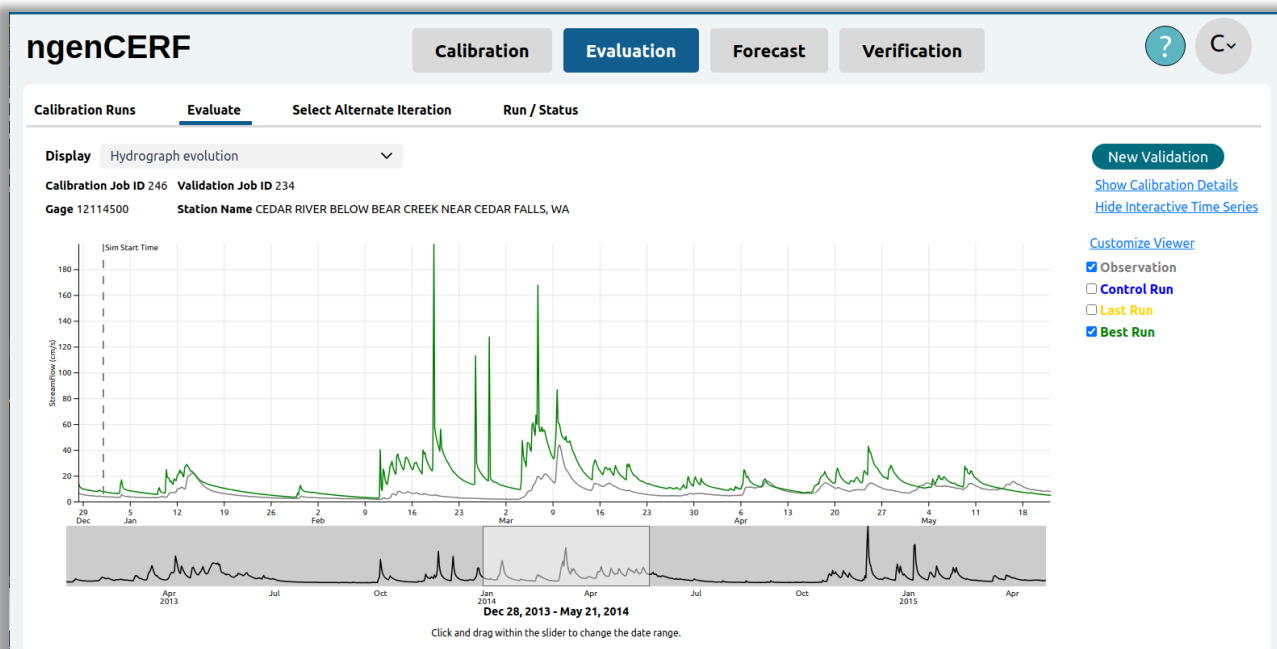


Figure 29 Evaluation → Evaluate Tab View: Time Series

5.5.3 Compare Permutations View

This tab allows the user to make a comparison of different calibration permutation runs.

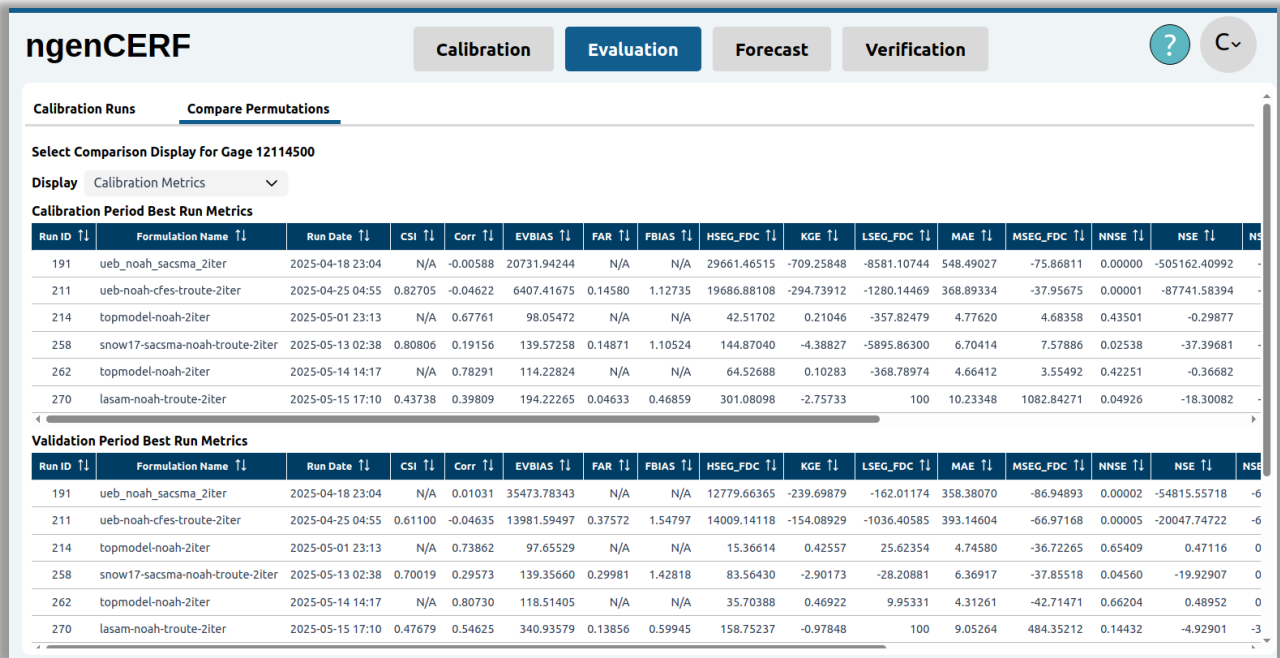


Figure 30 Evaluation → Evaluate Tab View: Compare Permutations

5.5.4 Select Alternative Iteration Tab

The **Select Alternate Iteration** tab is activated when the **New Validation Run** action is selected. This tab displays two tables stacked vertically: the upper table shows iteration metrics, including objective function values, and the lower table shows the corresponding parameter sets used for each iteration.

The tables are **synchronized** so that the same iterations remain aligned and visible in both tables, even when sorting is applied to columns in either table. To select an alternate parameter set for validation, the user can select the iteration row in either table then click Next to review the selection before running the validation using that iteration's parameters.

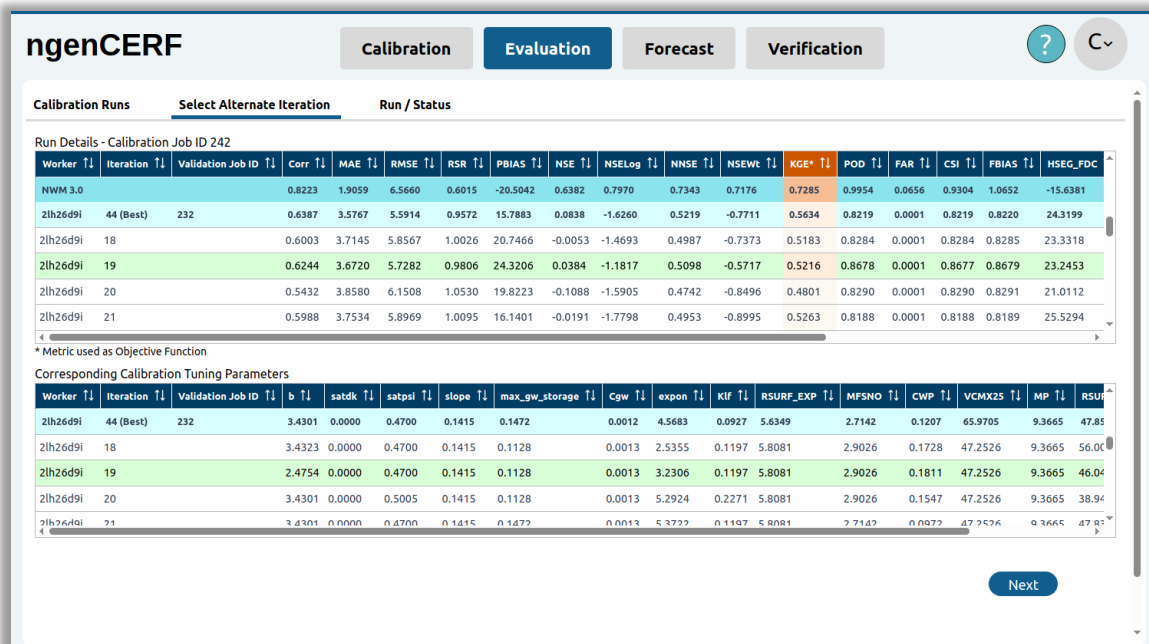


Figure 31 Evaluation → Select Alternate Iteration Tab

5.5.5 Run / Status Tab

This tab allows the user to start the validation run and monitor its progress while running. Once the run is complete, the **Evaluate** button becomes available. Click it to view and evaluate the results of the validation run.

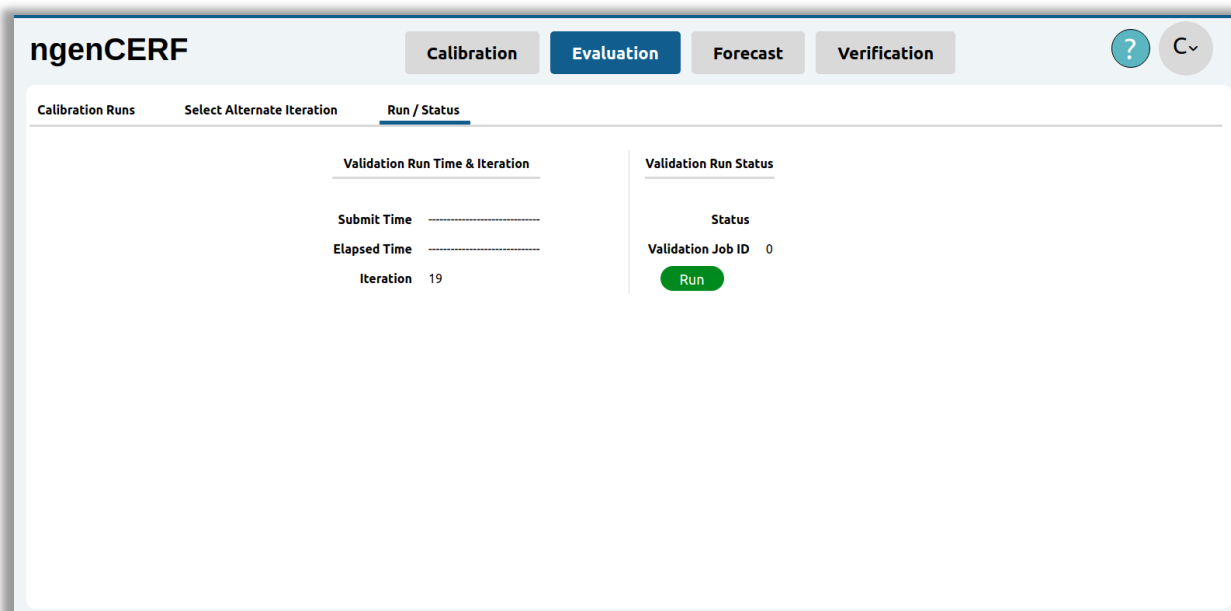
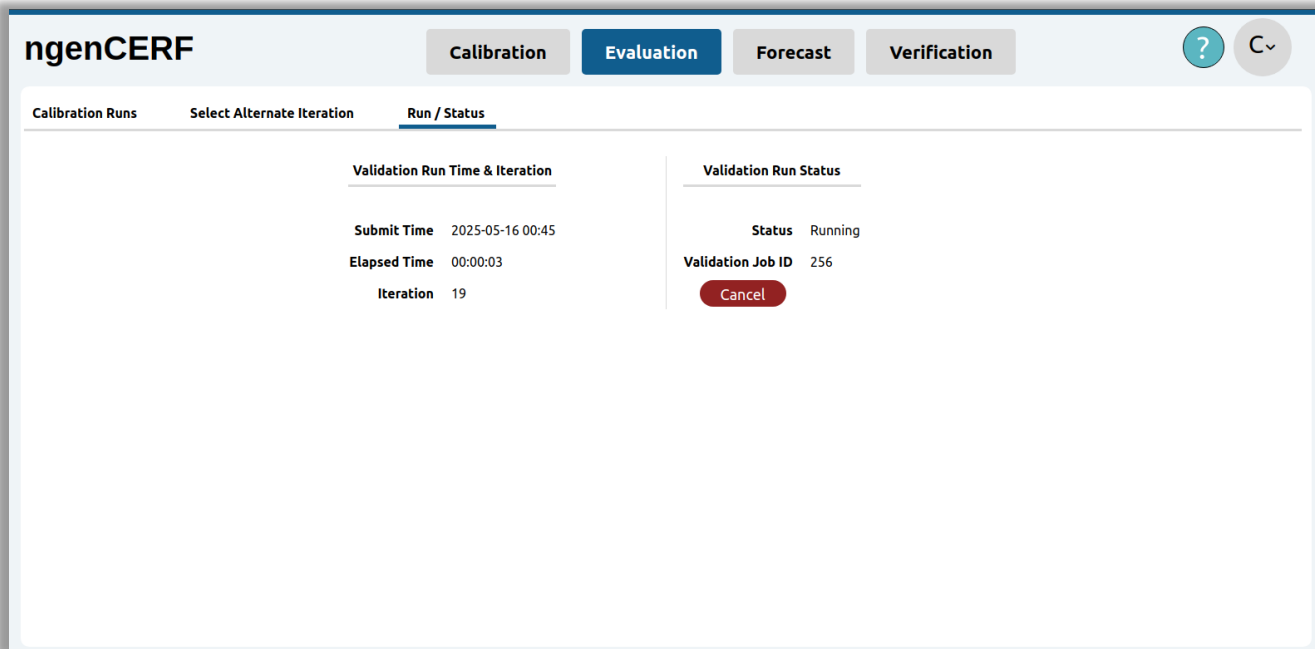


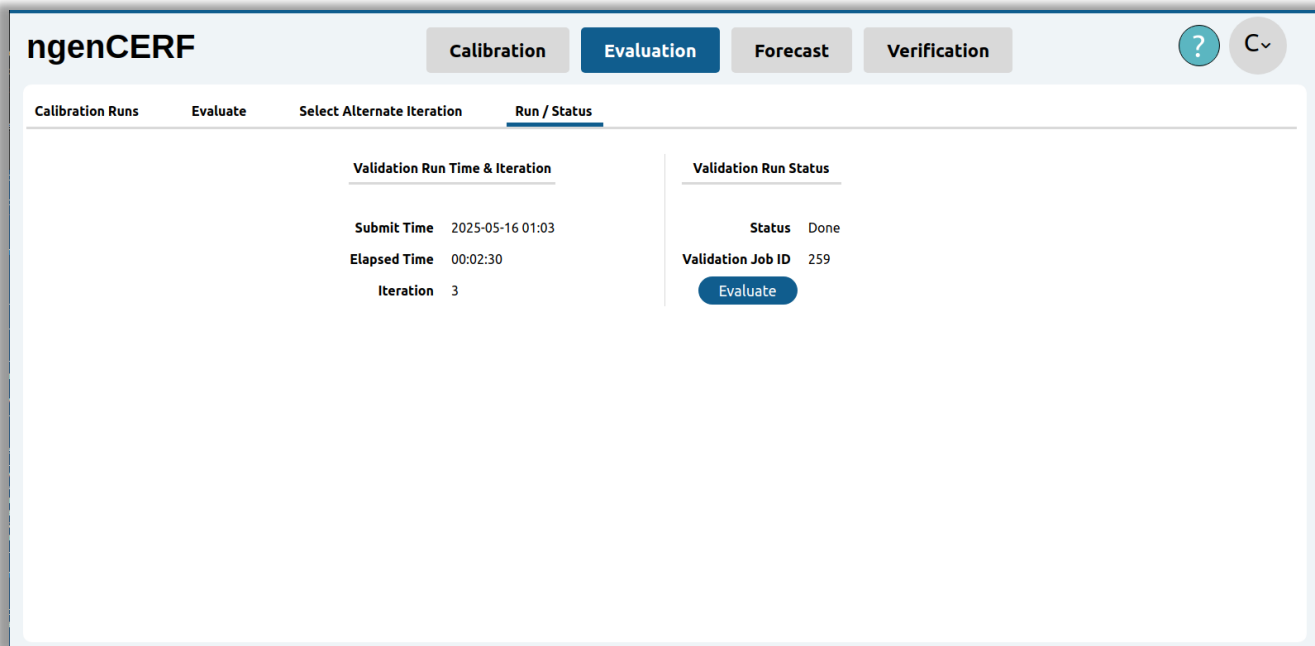
Figure 32 Evaluation → Run / Status Tab



The screenshot shows the ngenCERF interface with the 'Evaluation' tab selected. The 'Run / Status' sub-tab is active, displaying validation run details for Job ID 256. The status is 'Running'. The interface includes a 'Cancel' button to stop the run.

Validation Run Time & Iteration		Validation Run Status	
Submit Time	2025-05-16 00:45	Status	Running
Elapsed Time	00:00:03	Validation Job ID	256
Iteration	19		<button>Cancel</button>

Figure 33 Evaluation → Run / Status Tab View: Running



The screenshot shows the ngenCERF interface with the 'Evaluation' tab selected. The 'Run / Status' sub-tab is active, displaying validation run details for Job ID 259. The status is 'Done'. The interface includes an 'Evaluate' button to proceed to the next step.

Validation Run Time & Iteration		Validation Run Status	
Submit Time	2025-05-16 01:03	Status	Done
Elapsed Time	00:02:30	Validation Job ID	259
Iteration	3		<button>Evaluate</button>

Figure 34 Evaluation → Run / Status Tab View: Done

User Actions:

Run

Click to start the Validation Run

Cancel

Click to cancel the Validation Run

Evaluate

Go to the Evaluate tab

5.6 Forecast Workflow

The Forecast workflow uses the best parameter set produced during calibration to perform a user-selected forecast. ngenCERF currently supports eight distinct forecast types, allowing users to tailor forecast execution to their specific modeling objectives. Each forecast run is retained and may be individually selected for further analysis..

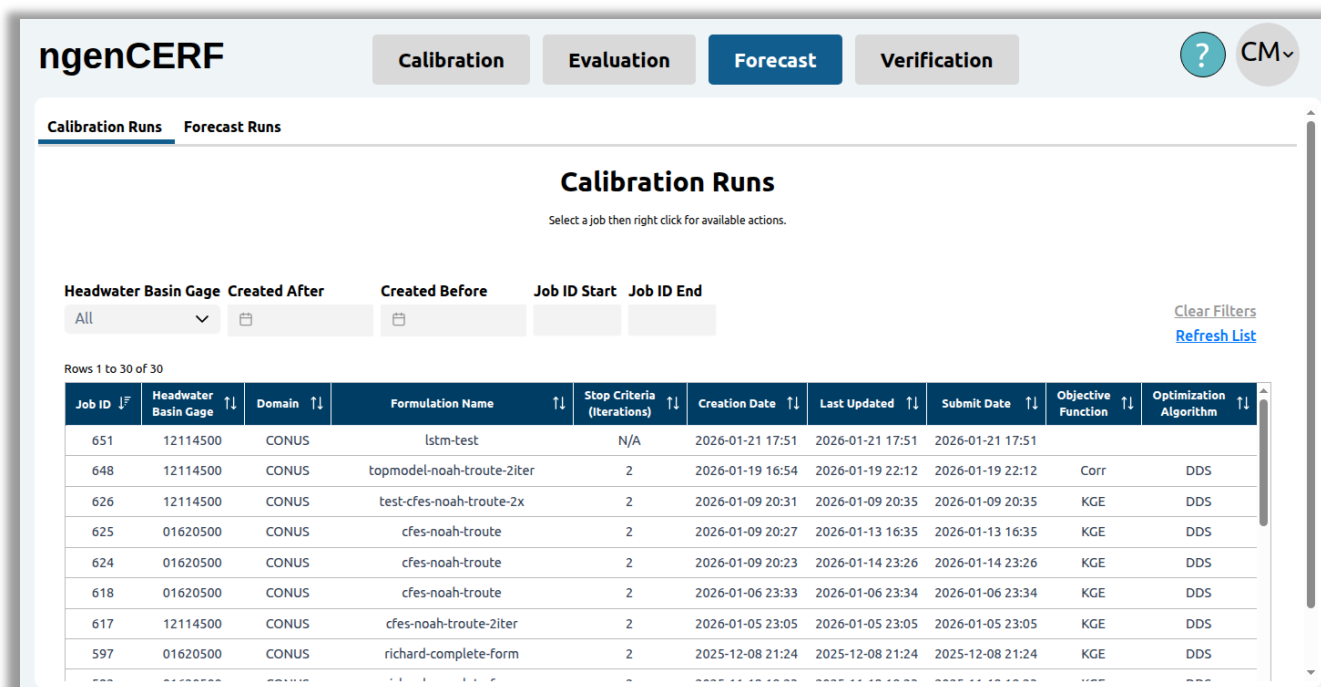
5.6.1 Calibration Runs Tab

This tab displays a table of completed calibration jobs that can be used to start new forecast runs. Users can select a calibration job and right-click to access available actions, including running a new forecast, viewing calibration details, downloading results, exporting the calibration configuration, deleting, or archiving the job.

Filters at the top of the table allow users to narrow the list of calibration jobs by Headwater Basin Gage, creation date, or Job ID range. Multiple filters can be applied simultaneously, and the table updates automatically. Use Clear Filters to reset all filters and Refresh List to update the table.

Archived calibration jobs do not appear in the table and cannot be used directly for forecasts. To use an archived job, it must first be unarchived in the Calibration workflow.

The table columns can be sorted by clicking the sort icon in the header.



The screenshot shows the ngenCERF interface with the Forecast tab selected. The Calibration Runs tab is active, displaying a table of calibration jobs. The table has columns for Job ID, Headwater Basin Gage, Domain, Formulation Name, Stop Criteria (Iterations), Creation Date, Last Updated, Submit Date, Objective Function, and Optimization Algorithm. The table is sorted by Job ID in descending order. There are 30 rows displayed, with the first row being Job ID 651 and the last row being Job ID 597. The table is titled 'Calibration Runs' and includes a note: 'Select a job then right click for available actions.' There are also filter buttons for Headwater Basin Gage, Created After, Created Before, Job ID Start, and Job ID End. The 'Clear Filters' and 'Refresh List' buttons are visible on the right side of the table.

Job ID	Headwater Basin Gage	Domain	Formulation Name	Stop Criteria (Iterations)	Creation Date	Last Updated	Submit Date	Objective Function	Optimization Algorithm
651	12114500	CONUS	lstm-test	N/A	2026-01-21 17:51	2026-01-21 17:51	2026-01-21 17:51		
648	12114500	CONUS	topmodel-noah-troute-2lter	2	2026-01-19 16:54	2026-01-19 22:12	2026-01-19 22:12	Corr	DDS
626	12114500	CONUS	test-cfes-noah-troute-2x	2	2026-01-09 20:31	2026-01-09 20:35	2026-01-09 20:35	KGE	DDS
625	01620500	CONUS	cfes-noah-troute	2	2026-01-09 20:27	2026-01-13 16:35	2026-01-13 16:35	KGE	DDS
624	01620500	CONUS	cfes-noah-troute	2	2026-01-09 20:23	2026-01-14 23:26	2026-01-14 23:26	KGE	DDS
618	01620500	CONUS	cfes-noah-troute	2	2026-01-06 23:33	2026-01-06 23:34	2026-01-06 23:34	KGE	DDS
617	12114500	CONUS	cfes-noah-troute-2lter	2	2026-01-05 23:05	2026-01-05 23:05	2026-01-05 23:05	KGE	DDS
597	01620500	CONUS	richard-complete-form	2	2025-12-08 21:24	2025-12-08 21:24	2025-12-08 21:24	KGE	DDS

Figure 35 Forecast → Calibration Runs Tab

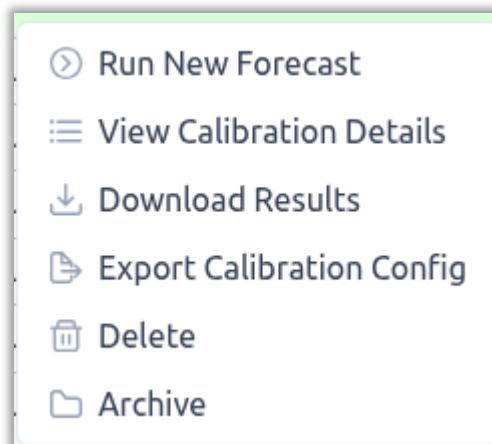


Figure 36 Forecast → Calibration Runs Selected Job Context Menu

Right Click Actions

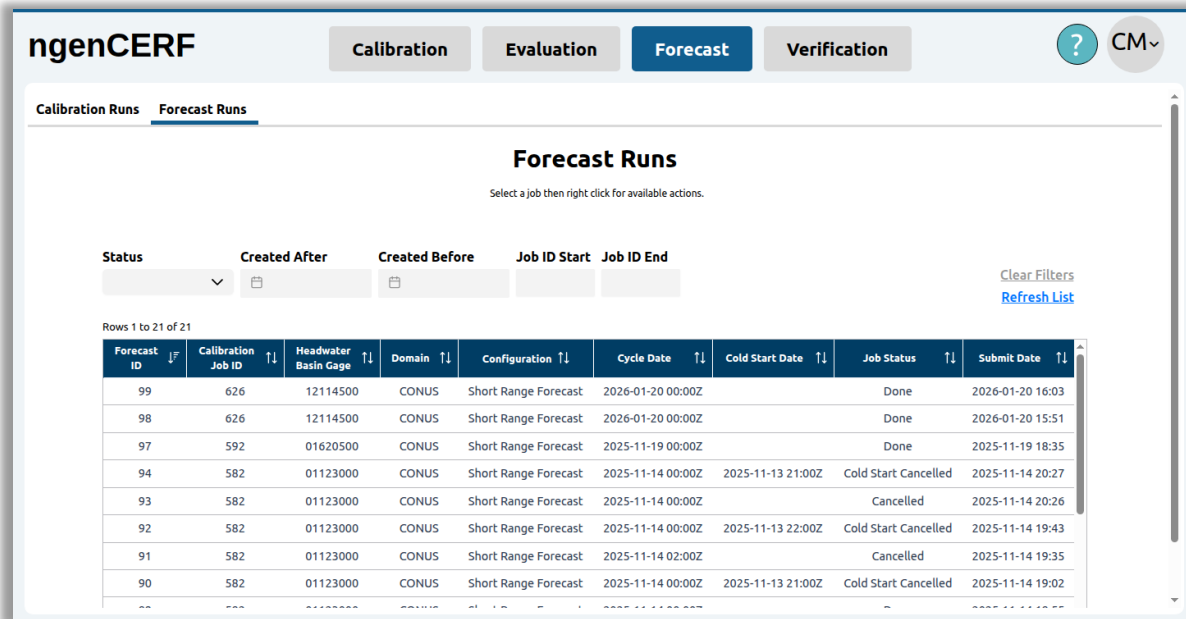
Run New Forecast	Run a new forecast based on the selected calibration job.
View Calibration Details	View the summary details of the calibration job.
Download Results	Download all data associated with a Done calibration job.
Export Calibration Config	Export the Calibration Definition File. (Used by the CLI.)
Delete	Permanently deletes the selected calibration job(s) from the system.
Archive	Archive the selected calibration job. Archived jobs will no longer appear in the Calibration Runs table. To use an archived job, it must first be unarchived from the Calibration workflow.

5.6.2 Forecast Runs Tab

The **Forecast Runs** tab displays a table of previous forecast runs with basic details, allowing users to review past jobs or start new forecasts. Users can filter the table by Headwater Basin Gage, Creation date range, or Job ID range. Multiple filters can be applied together, and the table updates automatically. Use Clear Filters to reset all filters and Refresh List to update the table.

Right-click actions for selected forecast runs include: viewing the job status or results, running a new forecast using the same calibration, viewing calibration details, and deleting the job. Deleted forecast runs are permanently removed along with all associated data.

Table columns can be sorted by clicking the sort icon in each column header.



ngenCERF Calibration Evaluation **Forecast** Verification ? CM~

Calibration Runs **Forecast Runs**

Forecast Runs
Select a job then right click for available actions.

Status Created After Created Before Job ID Start Job ID End [Clear Filters](#) [Refresh List](#)

Rows 1 to 21 of 21

Forecast ID	Calibration Job ID	Headwater Basin Gage	Domain	Configuration	Cycle Date	Cold Start Date	Job Status	Submit Date
99	626	12114500	CONUS	Short Range Forecast	2026-01-20 00:00Z		Done	2026-01-20 16:03
98	626	12114500	CONUS	Short Range Forecast	2026-01-20 00:00Z		Done	2026-01-20 15:51
97	592	01620500	CONUS	Short Range Forecast	2025-11-19 00:00Z		Done	2025-11-19 18:35
94	582	01123000	CONUS	Short Range Forecast	2025-11-14 00:00Z	2025-11-13 21:00Z	Cold Start Cancelled	2025-11-14 20:27
93	582	01123000	CONUS	Short Range Forecast	2025-11-14 00:00Z		Cancelled	2025-11-14 20:26
92	582	01123000	CONUS	Short Range Forecast	2025-11-14 00:00Z	2025-11-13 22:00Z	Cold Start Cancelled	2025-11-14 19:43
91	582	01123000	CONUS	Short Range Forecast	2025-11-14 02:00Z		Cancelled	2025-11-14 19:35
90	582	01123000	CONUS	Short Range Forecast	2025-11-14 00:00Z	2025-11-13 21:00Z	Cold Start Cancelled	2025-11-14 19:02

Figure 37 Forecast → Forecast Runs Tab

Filters

Use the filters at the top of the table to narrow down the list of forecast jobs. You can use any combination of filters; only jobs that match all active filters will be shown. The table updates automatically as you select filters.

Headwater Basin Gage

Select a single gage to show jobs for that basin.

Created After

Show jobs created on or after the selected date.

Created Before

Show jobs created on or before the selected date.

Job ID Start

Show jobs with ID greater than or equal to this number.

Job ID End

Show jobs with ID less than or equal to this number.

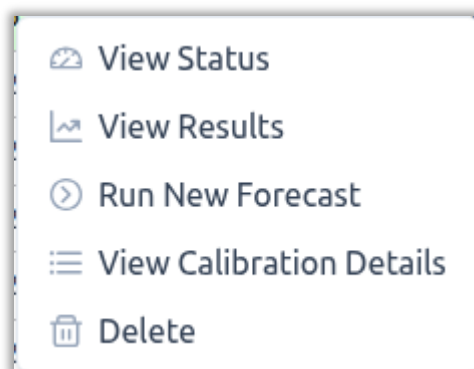


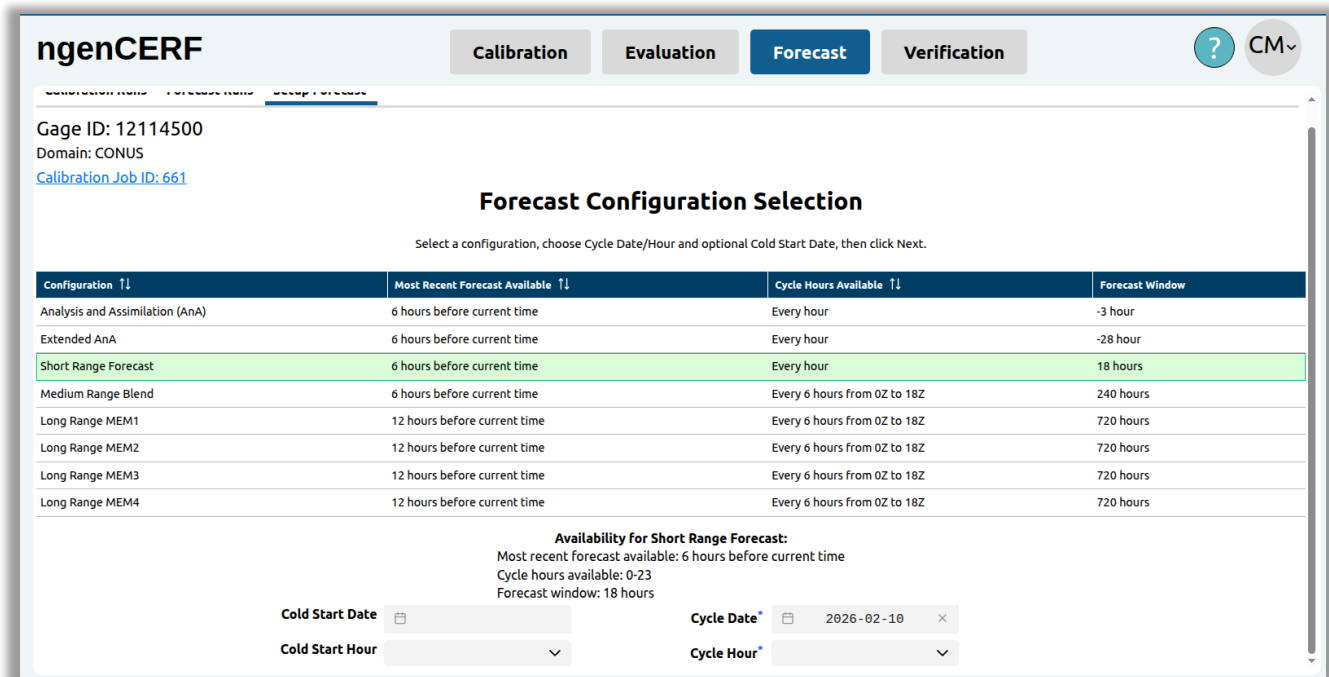
Figure 38 Forecast → Forecast Runs Selected Job Context Menu

Right Click Actions

View Status	View status of a running or completed forecast
View Results	View results of a successfully completed forecast
Run New Forecast	Run a new forecast based on the selected calibration job.
View Calibration Details	View the summary details of the calibration job.
Delete	Delete the forecast job.

5.6.3 Setup Forecast Tab

This tab allows the user to configure a new forecast run. Select a forecast configuration from the table, then enter the required input fields: Cycle Date and Cycle Hour (required), and optionally Cold Start Date and Cold Start Hour. Once all required fields are provided, the Next button becomes available to proceed to review and run the forecast.



ngenCERF Calibration Evaluation **Forecast** Verification ? CM

Gage ID: 12114500
 Domain: CONUS
[Calibration Job ID: 661](#)

Forecast Configuration Selection

Select a configuration, choose Cycle Date/Hour and optional Cold Start Date, then click Next.

Configuration ↑↓	Most Recent Forecast Available ↑↓	Cycle Hours Available ↑↓	Forecast Window
Analysis and Assimilation (AnA)	6 hours before current time	Every hour	-3 hour
Extended AnA	6 hours before current time	Every hour	-28 hour
Short Range Forecast	6 hours before current time	Every hour	18 hours
Medium Range Blend	6 hours before current time	Every 6 hours from 0Z to 18Z	240 hours
Long Range MEM1	12 hours before current time	Every 6 hours from 0Z to 18Z	720 hours
Long Range MEM2	12 hours before current time	Every 6 hours from 0Z to 18Z	720 hours
Long Range MEM3	12 hours before current time	Every 6 hours from 0Z to 18Z	720 hours
Long Range MEM4	12 hours before current time	Every 6 hours from 0Z to 18Z	720 hours

Availability for Short Range Forecast:
 Most recent forecast available: 6 hours before current time
 Cycle hours available: 0-23
 Forecast window: 18 hours

Cold Start Date Cycle Date*

Cold Start Hour Cycle Hour*

Figure 39 Forecast → Setup Forecast

User Actions:

Configuration	Type of Forecast configuration ran for this job.
Cold Start Date	Optional cold start date used to initialize the forecast and must be earlier than the Cycle Date .
Cold Start Hour	Optional cold start hour used to initialize the forecast and must be earlier than the Cycle Date .

Cycle Start Date*

The cycle date defines the forecast initialization date and controls the selection of real-time forcing data used to generate the forecast.

Cycle Start Hour*

The cycle hour defines the forecast initialization hour and controls the selection of real-time forcing data used to generate the forecast.

* An asterisk indicates a required field

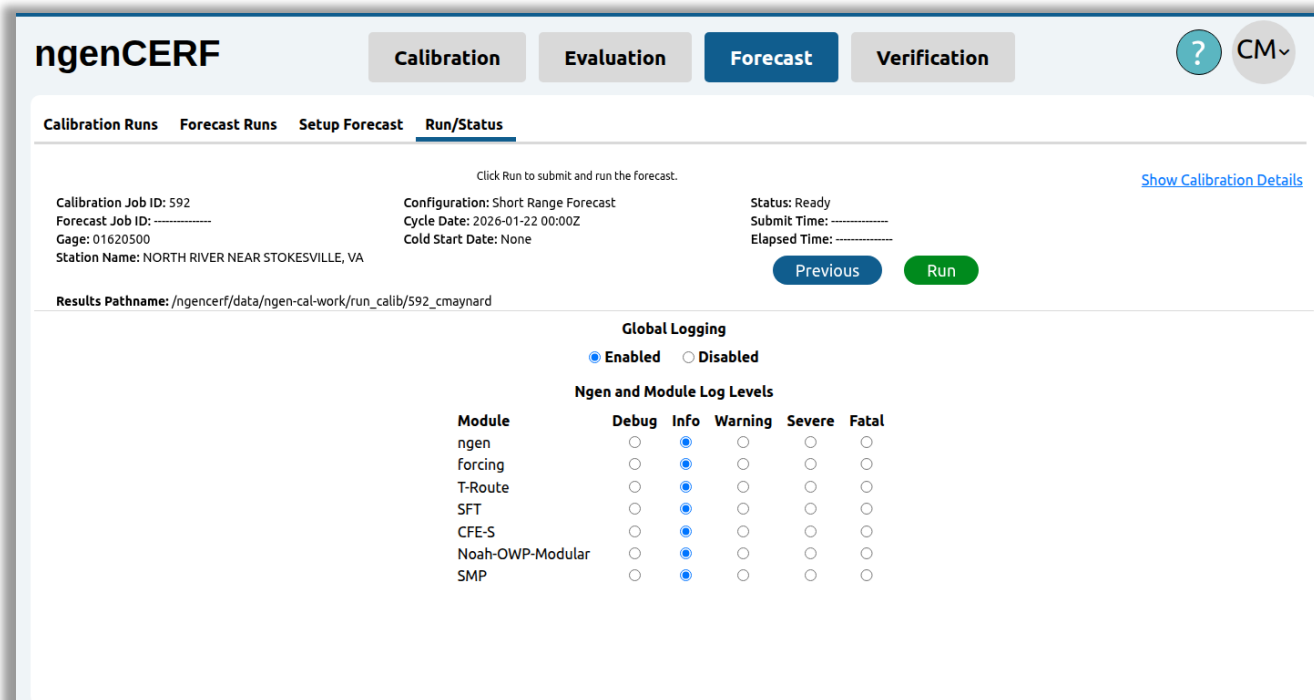
5.6.4 Status / Run Tab

The Status / Run tab allows the user to execute a forecast and monitor its progress. When the **Run** button is clicked, the latest real-time forcing data is downloaded (which may take several minutes), after which the forecast automatically starts. The tab displays key job details including the calibration job used as the basis, forecast job ID, submit time, elapsed time, status, cycle and cold start dates, configuration, and results pathname.

Log files may not be immediately available. If the selected item is unavailable, a “Log file unavailable” notice will appear. Once the log is ready, selecting the item from the Display dropdown will show it.

While the forecast is running, the **Cancel** button can be used to stop the job. Once the job is complete, the View Results button becomes available to review the forecast outputs. Global logging and log levels can be configured as needed.

Navigation to the previous tab is available via the **Previous** button until the forecast is started.



The screenshot shows the ngenCERF web interface with the 'Forecast' tab selected. The 'Run/Status' sub-tab is active, displaying job details and logging options.

ngenCERF | Calibration | Evaluation | **Forecast** | Verification | ? CM~

Calibration Runs | Forecast Runs | Setup Forecast | **Run/Status**

Click Run to submit and run the forecast. [Show Calibration Details](#)

Calibration Job ID: 592
 Forecast Job ID: -----
 Gage: 01620500
 Station Name: NORTH RIVER NEAR STOKESVILLE, VA

Configuration: Short Range Forecast
 Cycle Date: 2026-01-22 00:00Z
 Cold Start Date: None

Status: Ready
 Submit Time: -----
 Elapsed Time: -----

[Previous](#) [Run](#)

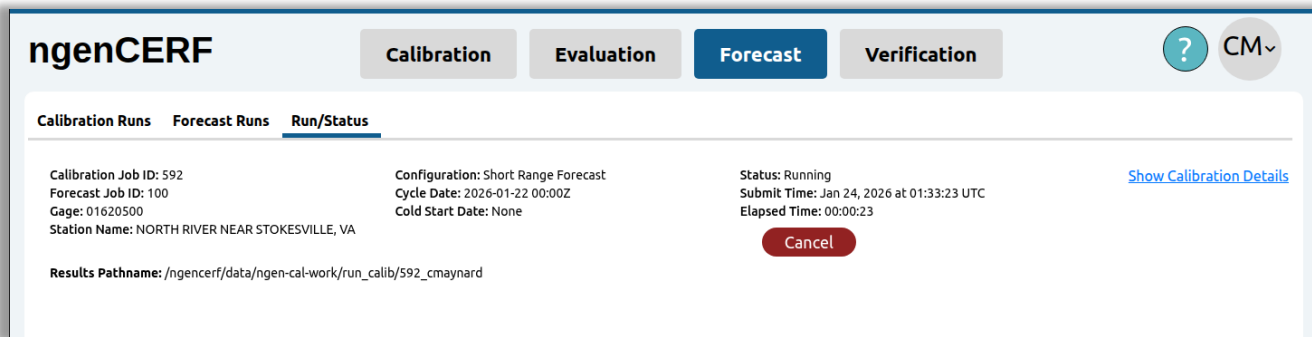
Results Pathname: /ngencerf/data/ngen-cal-work/run_calib/592_cmaynard

Global Logging
☒ Enabled ☐ Disabled

Ngem and Module Log Levels

Module	Debug	Info	Warning	Severe	Fatal
ngen	☐	☒	☐	☐	☐
forcing	☐	☒	☐	☐	☐
T-Route	☐	☒	☐	☐	☐
SFT	☐	☒	☐	☐	☐
CFE-S	☐	☒	☐	☐	☐
Noah-OWP-Modular	☐	☒	☐	☐	☐
SMP	☐	☒	☐	☐	☐

Figure 40 Forecast → Status / Run Tab



ngenCERF Calibration Evaluation **Forecast** Verification ? CM

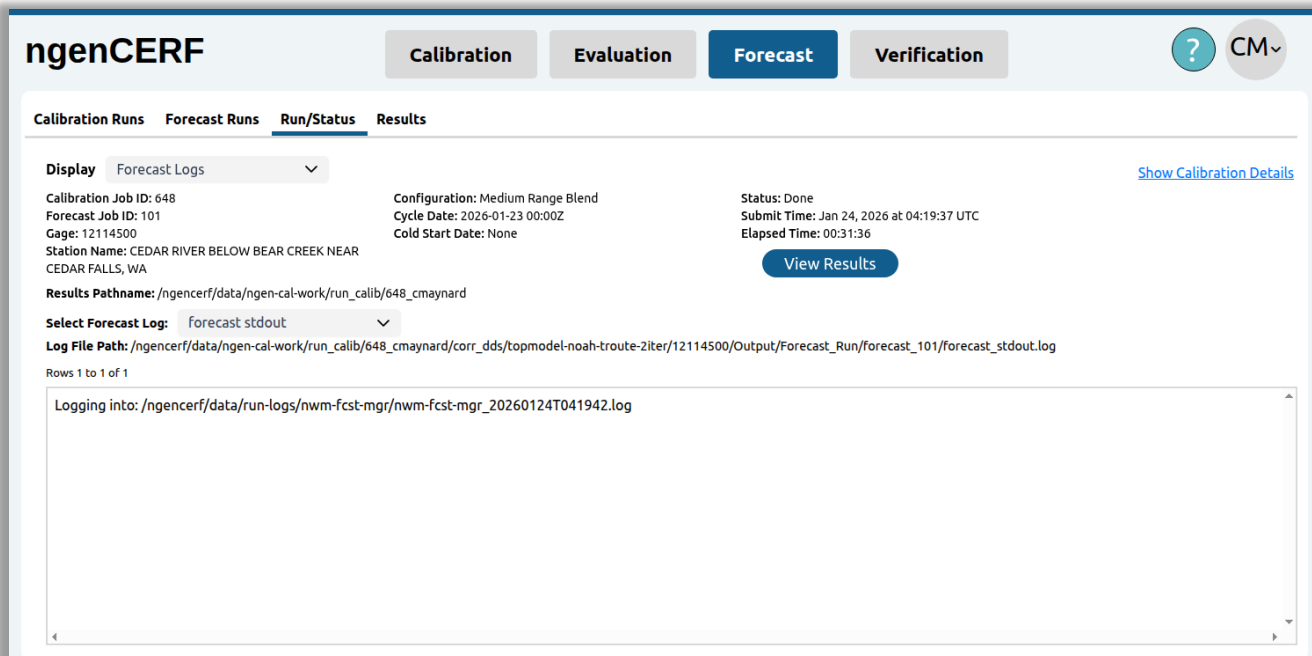
Calibration Runs Forecast Runs **Run/Status**

Calibration Job ID: 592 Configuration: Short Range Forecast Status: Running
 Forecast Job ID: 100 Cycle Date: 2026-01-22 00:00Z Submit Time: Jan 24, 2026 at 01:33:23 UTC
 Gage: 01620500 Cold Start Date: None Elapsed Time: 00:00:23
 Station Name: NORTH RIVER NEAR STOKESVILLE, VA
 Results Pathname: /ngencerf/data/ngen-cal-work/run_calib/592_cmaynard

[Show Calibration Details](#)

Cancel

Figure 41 Forecast → Status / Run Tab View: Running



ngenCERF Calibration Evaluation **Forecast** Verification ? CM

Calibration Runs Forecast Runs **Run/Status** Results

Display Forecast Logs

Calibration Job ID: 648 Configuration: Medium Range Blend Status: Done
 Forecast Job ID: 101 Cycle Date: 2026-01-23 00:00Z Submit Time: Jan 24, 2026 at 04:19:37 UTC
 Gage: 12114500 Cold Start Date: None Elapsed Time: 00:31:36
 Station Name: CEDAR RIVER BELOW BEAR CREEK NEAR CEDAR FALLS, WA
 Results Pathname: /ngencerf/data/ngen-cal-work/run_calib/648_cmaynard
 Select Forecast Log: forecast stdout
 Log File Path: /ngencerf/data/ngen-cal-work/run_calib/648_cmaynard/corr_dds/topmodel-noah-trout-2liter/12114500/Output/Forecast_Run/forecast_101/forecast_stdout.log

[Show Calibration Details](#)

View Results

Rows 1 to 1 of 1

Logging into: /ngencerf/data/run-logs/nwm-fcst-mgr/nwm-fcst-mgr_20260124T041942.log

Figure 42 Forecast → Status / Run Tab View: Done

User Actions:

Display

Use this dropdown to choose the result to view, including time-series plots or logs.

- Selecting **Logs** reveals a second pull-down to select a specific log.
- Selecting **Time Series** shows a slider at the bottom to zoom in or out. The plot and slider default to fully zoomed out.

Calibration Job ID

Calibration Job ID use as the basis for this forecast.

Forecast Job ID

This forecast job ID.

Gage

Gaged used for this forecast.

Location Name

Location name for the gage.

Results Pathname

File system location containing all data for this job.

Configuration

Type of Forecast configuration ran for this job.

Cold Start Date

Optional cold start date and hour used to initialize the forecast and must be earlier than the **Cycle Date**.

Cycle Date

The forecast cycle date and hour which controls the selection of real-time forcing data used to generate the forecast.

Status

Status of the completed job.

Submit Time

UTC Date/Time the job was submitted to run.

Elapsed Time

Total time the job took to run, displayed as days, hours, minutes, and seconds (DD:HH:MM:SS).

5.6.5 Results Tab

After the forecast is done, view any auto-generated output, including logs, by selecting the item from the Display dropdown.

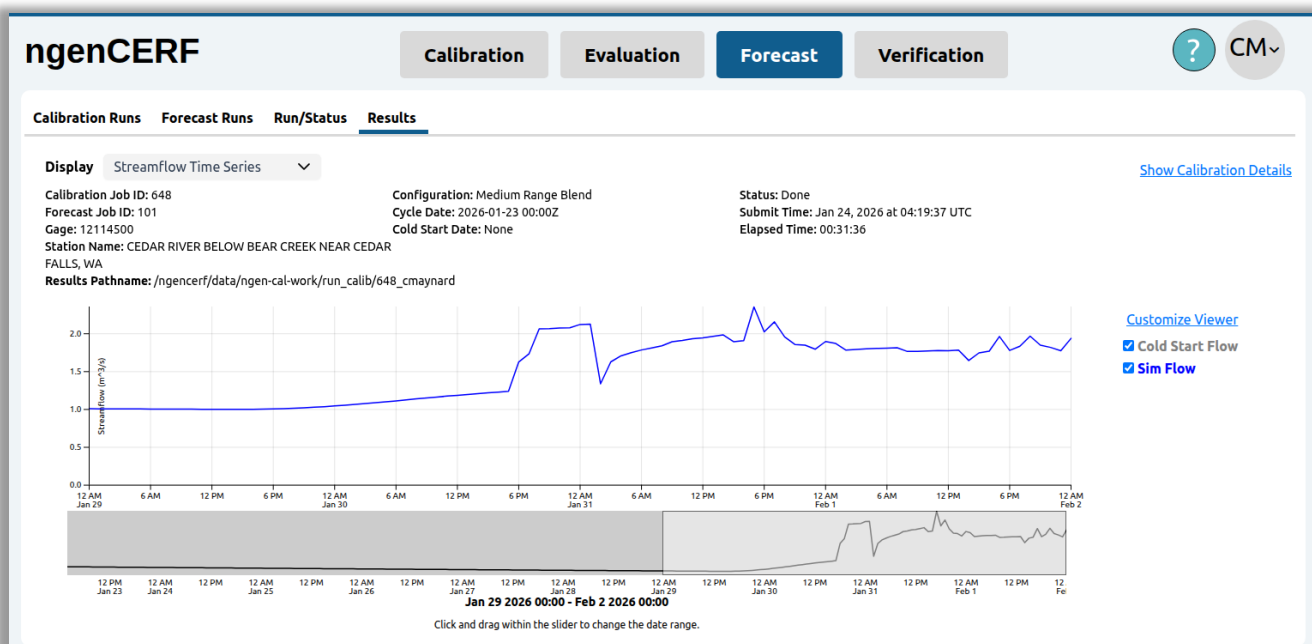


Figure 43 Forecast → Results Tab

User Actions:

Display

Use this dropdown to choose the result to view, including time-series plots or logs.

- Selecting **Logs** reveals a second pull-down to select a specific log.

- Selecting **Time Series** shows a slider at the bottom to zoom in or out. The plot and slider default to fully zoomed out.

Calibration Job ID	Calibration Job ID use as the basis for this forecast.
Forecast Job ID	This forecast job ID.
Gage	Gaged used for this forecast.
Location Name	Location name for the gage.
Results Pathname	File system location containing all data for this job.
Configuration	Type of Forecast configuration ran for this job.
Cold Start Date	Optional cold start date and hour used to initialize the forecast and must be earlier than the Cycle Date .
Cycle Date	The forecast cycle date and hour which controls the selection of real-time forcing data used to generate the forecast.
Status	Status of the completed job.
Submit Time	UTC Date/Time the job was submitted to run.
Elapsed Time	Total time the job took to run, displayed as days, hours, minutes, and seconds (DD:HH:MM:SS).

5.7 Verification Workflow

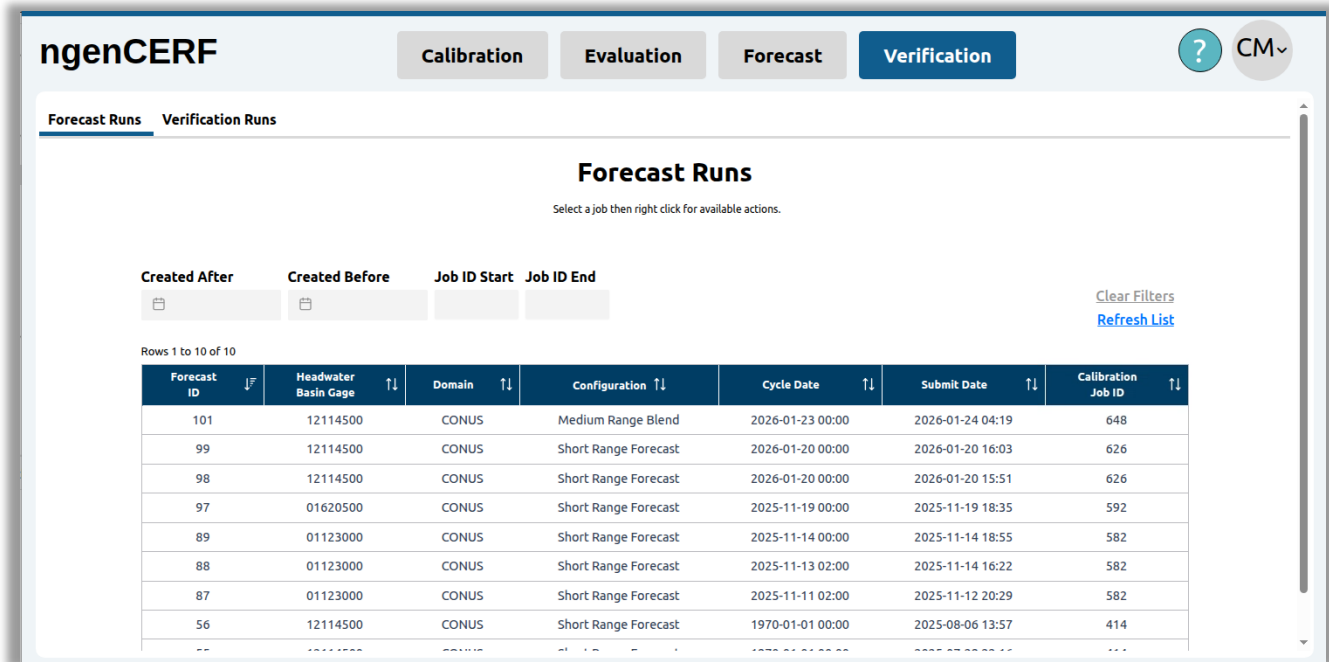
Verification compares forecast outputs against corresponding observational or reference datasets, enabling users to assess forecast performance for a specific forecast configuration.

5.7.1 Forecast Runs Tab

The **Forecast Runs** tab displays a table of previous forecast runs with basic details, allowing users to select a job to use the basis for a verification run. Users can filter the table by Status, Creation date range, or Job ID range. Multiple filters can be applied together, and the table updates automatically. Use Clear Filters to reset all filters and Refresh List to update the table.

Right-click action for a forecast run allows the user to **run** the verification.

Table columns can be sorted by clicking the sort icon in each column header.



ngenCERF Calibration Evaluation Forecast **Verification** ? CM

Forecast Runs Verification Runs

Forecast Runs
Select a job then right click for available actions.

Created After Created Before Job ID Start Job ID End

[Clear Filters](#)
[Refresh List](#)

Rows 1 to 10 of 10

Forecast ID	Headwater Basin Gage	Domain	Configuration	Cycle Date	Submit Date	Calibration Job ID
101	12114500	CONUS	Medium Range Blend	2026-01-23 00:00	2026-01-24 04:19	648
99	12114500	CONUS	Short Range Forecast	2026-01-20 00:00	2026-01-20 16:03	626
98	12114500	CONUS	Short Range Forecast	2026-01-20 00:00	2026-01-20 15:51	626
97	01620500	CONUS	Short Range Forecast	2025-11-19 00:00	2025-11-19 18:35	592
89	01123000	CONUS	Short Range Forecast	2025-11-14 00:00	2025-11-14 18:55	582
88	01123000	CONUS	Short Range Forecast	2025-11-13 02:00	2025-11-14 16:22	582
87	01123000	CONUS	Short Range Forecast	2025-11-11 02:00	2025-11-12 20:29	582
56	12114500	CONUS	Short Range Forecast	1970-01-01 00:00	2025-08-06 13:57	414

Figure 44 Verification → Forecast Runs Tab

Filters

Use the filters at the top of the table to narrow down the list of forecast jobs. You can use any combination of filters; only jobs that match all active filters will be shown. The table updates automatically as you select filters.

Created After	Show jobs created on or after the selected date.
Created Before	Show jobs created on or before the selected date.
Job ID Start	Show jobs with ID greater than or equal to this number.
Job ID End	Show jobs with ID less than or equal to this number.

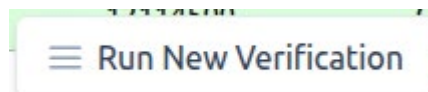


Figure 45 Verification → Forecast Runs Selected Job Context Menu

Right Click Actions

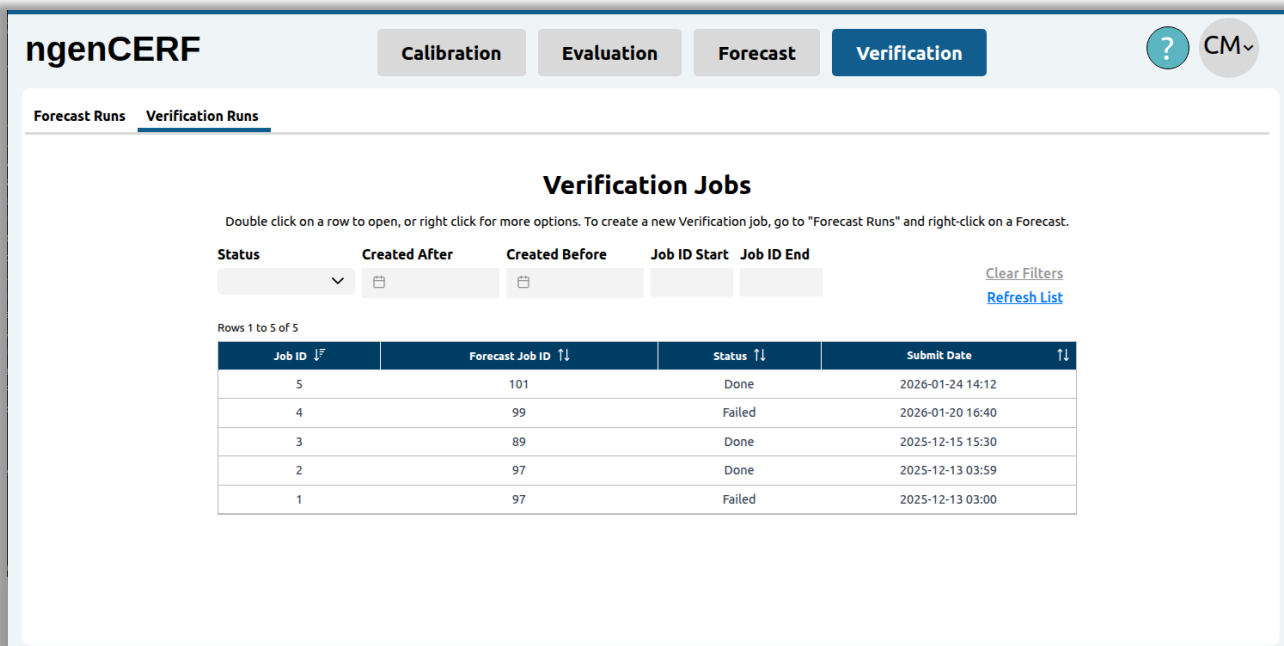
Run New Verification	Run a new verification based on the selected forecast job.
-----------------------------	--

5.7.2 Verification Runs Tab

The **Verification Runs** tab displays a table of previous verification runs with basic details, allowing users to select a job to use the basis for a verification run. Users can filter the table by Status, Creation date range, or Job ID range. Multiple filters can be applied together, and the table updates automatically. Use Clear Filters to reset all filters and Refresh List to update the table.

Right clicking a verification run allows you to view the job's status, examine results of a completed verification, or delete the job.

Table columns can be sorted by clicking the sort icon in each column header.



ngenCERF Calibration Evaluation Forecast **Verification** ? CM

Forecast Runs **Verification Runs**

Verification Jobs

Double click on a row to open, or right click for more options. To create a new Verification job, go to "Forecast Runs" and right-click on a Forecast.

Status Created After Created Before Job ID Start Job ID End [Clear Filters](#)
[Refresh List](#)

Rows 1 to 5 of 5

Job ID ↑↓	Forecast Job ID ↑↓	Status ↑↓	Submit Date ↑↓
5	101	Done	2026-01-24 14:12
4	99	Failed	2026-01-20 16:40
3	89	Done	2025-12-15 15:30
2	97	Done	2025-12-13 03:59
1	97	Failed	2025-12-13 03:00

Figure 46 Verification → Verification Runs Tab

Filters

Use the filters at the top of the table to narrow down the list of forecast jobs. You can use any combination of filters; only jobs that match all active filters will be shown. The table updates automatically as you select filters.

Created After	Show jobs created on or after the selected date.
Created Before	Show jobs created on or before the selected date.
Job ID Start	Show jobs with ID greater than or equal to this number.
Job ID End	Show jobs with ID less than or equal to this number.

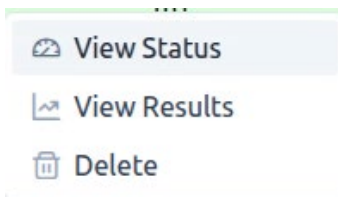


Figure 47 Verification → Verification Runs Selected Job Context Menu

Right Click Actions

View Status	View status of a running or completed verification
View Results	View results of a successfully completed verification
Delete	Delete the verification job.

5.7.3 Status / Run Tab

The **Status / Run** tab allows the user to execute a verification and monitor its progress. The tab displays key job details including the forecast job used as the basis, verification job ID, forecast configuration, cycle date, status, submit time, elapsed time and results pathname. When the **Run** button is clicked the verification process.

Log files may not be immediately available. If the selected item is unavailable, a “Log file unavailable” notice will appear. Once the log is ready, selecting the item from the Display dropdown will show it.

While the verification is running, the **Cancel** button can be used to stop the job. Once the job is complete, the **View Results** button becomes available to review the verification outputs. Global logging and log levels can be configured as needed.

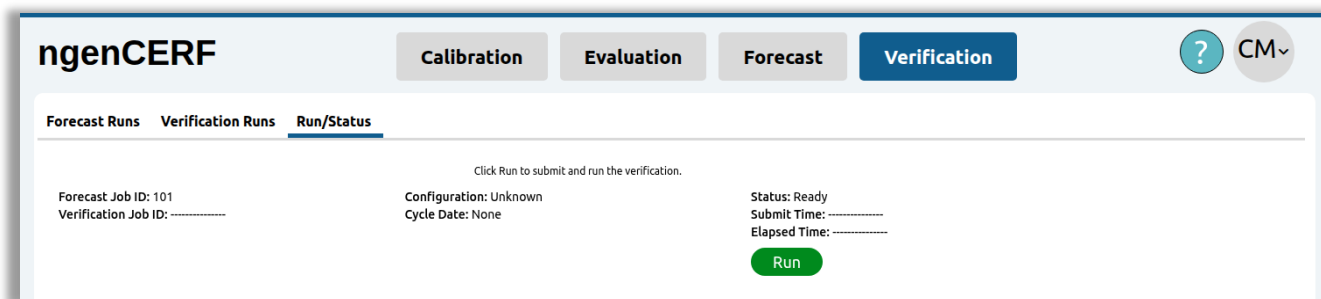


Figure 48 Verification → Status / Run Tab

ngenCERF

Calibration

Evaluation

Forecast

Verification

?

CM

Forecast Runs

Verification Runs

Run/Status

Display

Verification Logs

Forecast Job ID: 103

Configuration: Medium Range Blend

Status: Running

Verification Job ID: 9

Cycle Date: 2026-01-26 00:00 UTC

Submit Time: Jan 28, 2026 at 16:19:24 UTC

Elapsed Time: 00:00:11

Cancel

Select Verification Log:

verification stdout

Log File Path:

/ngencertif/data/ngen-cal-work/run_calib/648_cmaynard/corr_dds/topmodel-noah-troute-2iter/12114500/Output/Forecast_Run/forecast_103/Verification_Run/verification_9/verification_stdout.log

Log file unavailable

Figure 49 Verification → Status / Run Tab View: Running

ngenCERF

Calibration

Evaluation

Forecast

Verification

?

CM

Forecast Runs

Verification Runs

Run/Status

Results

Display

Verification Logs

Forecast Job ID: 101

Configuration: Medium Range Blend

Status: Done

Verification Job ID: 5

Cycle Date: 2026-01-23 00:00Z

Submit Time: Jan 24, 2026 at 14:12:34 UTC

Elapsed Time: 00:00:15

View Results

Select Verification Log:

verification stdout

Log File Path:

/ngencertif/data/ngen-cal-work/run_calib/648_cmaynard/corr_dds/topmodel-noah-troute-2iter/12114500/Output/Forecast_Run/forecast_101/Verification_Run/verification_5/verification_stdout.log

Rows 1 to 27 of 27

2026-01-24 14:12:45,409 - nwm.verf.fetch_data - [INFO] - topmodel-noah-troute-2iter - Number of gages with observation data available: 1

2026-01-24 14:12:45,409 - __main__ - [INFO] - Execution time for fetch_obs_data: 1.6544289588928223 seconds

2026-01-24 14:12:45,470 - nwm.verf.pair_data - [INFO] - Joining time series for dataset topmodel-noah-troute-2iter in DuckDBDatabase: /ngencertif/data/ngen-cal-work/run_calib/648_cmaynard/c

2026-01-24 14:12:45,556 - nwm.verf.pair_data - [INFO] - Exporting paired data to /ngencertif/data/ngen-cal-work/run_calib/648_cmaynard/corr_dds/topmodel-noah-troute-2iter/12114500/Output

2026-01-24 14:12:45,603 - __main__ - [INFO] - Execution time for pair_data: 0.1936185359954834 seconds

2026-01-24 14:12:45,603 - nwm.verf.calc_metrics - [INFO] - Metrics will be calculated using nwm.eval library

2026-01-24 14:12:45,604 - nwm.verf.calc_metrics - [INFO] - Calculating metrics for topmodel-noah-troute-2iter ...

2026-01-24 14:12:45,627 - nwm.verf.calc_metrics - [WARNING] - Missing lead times: [1.0, 38.0, 39.0, 40.0, 41.0, 42.0, 43.0, 44.0, 45.0, 46.0, 47.0, 48.0, 49.0, 50.0, 51.0, 52.0, 53.0, 54.0, 55.0, 56.0, 57.0]

2026-01-24 14:12:46,102 - __main__ - [INFO] - Execution time for compute_metrics: 0.4986841678619385 seconds

2026-01-24 14:12:46,406 - nwm.verf.create_plots - [INFO] - Time series plots created at: /ngencertif/data/ngen-cal-work/run_calib/648_cmaynard/corr_dds/topmodel-noah-troute-2iter/12114500/

2026-01-24 14:12:47,015 - nwm.verf.create_plots - [INFO] - Metric table plots created at: /ngencertif/data/ngen-cal-work/run_calib/648_cmaynard/corr_dds/topmodel-noah-troute-2iter/12114500/

2026-01-24 14:12:48,752 - nwm.verf.create_plots - [INFO] - Barchart plots created at: /ngencertif/data/ngen-cal-work/run_calib/648_cmaynard/corr_dds/topmodel-noah-troute-2iter/12114500/Ou

2026-01-24 14:12:48,752 - __main__ - [INFO] - Execution time for plot_metrics: 2.6497411727905273 seconds

Figure 50 Verification → Status / Run Tab View: Done

ngenCERF

Calibration

Evaluation

Forecast

Verification

?

CM

Forecast Runs

Verification Runs

Run/Status

Results

Display

Select an option

Forecast Job ID: 101

Configuration: Medium Range Blend

Status: Done

Verification Job ID: 7

Cycle Date: 2026-01-23 00:00 UTC

Submit Time: Jan 27, 2026 at 02:01:52 UTC

Elapsed Time: 00:00:26

Figure 51 Verification → Results Tab

User Actions:

Display	Use this dropdown to view available logs. Selecting Logs reveals a second dropdown to choose a specific log.
Forecast Job ID	Forecast Job ID used as the basis for this verification.
Verification Job ID	This verification job ID.
Configuration	Configuration of the forecast job.
Cycle Date	The forecast cycle date and hour.
Status	Status of the completed job.
Submit Time	UTC Date/Time the job was submitted to run.
Elapsed Time	Total time the job took to run, displayed as days, hours, minutes, and seconds (DD:HH:MM:SS).
Results Pathname	File system location containing all data for this job.

6.0 External Data

The backend will retrieve data from Data Services, which retrieves it from external sources, for use within the GUI and the ngen app.

Table 7 Calibration Data Sources

Data	Source Format	Ngen Format	Source
Forcing	netCDF or zarr	csv	NOAA AORC Dataset AWS S3 Bucket or Forcing Engine
Observational	Time series	csv	USGS National Water Information System
Subsetting Hydrofabric	gpkg	gpkg	NGWPC Hydrofabric Service API call

6.1 Forcing Data

The forcing data represents the forces affecting water input to a basin on an hourly basis (aka time series data). The following information is provided in the NOAA Analysis of Record for Calibration (AORC) dataset:

Table 8: Calibration Forcing Data Items

netCDF name	Zarr name*	Description*	Units
time	time	YYYY-MM-DD HH:00:00	
RAINRATE	APCP_surface	Hourly total precipitation	kg/m ²

T2D	TMP_2maboveground	Temperature at 2 m above ground level (AGL)	°K
Q2D	SPFH_2maboveground	Specific humidity at 2m AGL	kg/kg
PSFC	PRES_surface	Air pressure at surface	Pa
SWDOWN	DSWRF_surface	Downward shortwave (solar) radiation flux at the surface	W/m ²
LWDOWN	DLWRF_surface	Downward longwave (infrared) radiation flux at the surface	W/m ²
U2D	UGRD_10maboveground	U (west-east) component of the wind at 10 m AGL	m/s
V2D	VGRD_10maboveground	V (south-north) components of the wind at 10 m AGL	m/s

The ngen app expects the data to be in a comma delimited file for each headwater basement catchment associated with the gage.

Example: First few lines from aorc_nwm/cat-10617.csv

```
data > aorc_nwm > csv_basin_group1 > Gage_01123000 > cat-10617.csv > data
1 time,LWDOWN,PSFC,Q2D,RAINRATE,SWDOWN,T2D,U2D,V2D
2 2017-01-01 01:00:00,310.565436814698,99191.9671872943,0.0042,0.000112143136021398,0.277.3817030727,4.39607844384344,8.64363990515544
3 2017-01-01 02:00:00,304.057655192448,99123.7209400303,0.00426557767488709,0.000260330787711924,0.276.255007527034,4.50694542223108,8.41304797704547
4 2017-01-01 03:00:00,307.840746061239,98868.1567136485,0.00404472563367488,0.000180081722885704,0.275.901037957814,4.64419528157747,8.18862073657279
5 2017-01-01 04:00:00,301.00244040728,99031.0588623861,0.00400265617365005,0.0001580745066770506,0.274.6259610372101,4.98118912622364,6.94355216746747
```

Figure 52 External Data – Example: AORC Forcings Data

Observational Data

The observational data represents the streamflow in a time series at a gage.

For the ngen app, it is expected to be in a comma delimited file (CSV) associated with the gage.

Example: First few lines from streamflow_obs/01123000_hourly_discharge.csv

```
data > streamflow_obs > 01123000_hourly_discharge.csv > data
1 dateTime,q_cms
2 2017-01-01 05:00:00,1.4894661522
3 2017-01-01 06:00:00,1.4894661522
4 2017-01-01 07:00:00,1.4894661522
5 2017-01-01 08:00:00,1.4894661522
```

Figure 53 External Data – Example: Observational Data

6.2 Subsetted Hydrofabric

The backend will call Data Services to obtain the subsetted hydrofabric GeoPackage (gpkg) for the Calibration run. A GeoPackage is an open, standards-based, platform-independent, portable, self-describing, compact format for transferring geospatial information. A GeoPackage is the SQLite container, and it supports direct use. This means that data in a

GeoPackage can be accessed and updated in a “native” storage format without intermediate format translations⁵.

To view the GeoPackage file, use a program like QGIS.

6.3 Example Calibratable Parameters Load File

This file can be a CSV or text file and must have a header line with param, min, max, init, and model. Each line in the file identifies the calibratable parameter name, its minimum value, its maximum value, its initial value and the model name. This information can be comma separated, space separated, or tab separated. The parameter and model names are case-sensitive.

Valid model names are:

CFE-S, CFE-X, LASAM, Noah-OWP-Modular, Sac-SMA, Snow-17, TopModel, UEB.

Example csv file:

```
param,min,max,init,model
field_capacity,10.5,500.0,20.0,LASAM
hydraulic_conductivity,1.0,50.0,10.0,LASAM
smcmax,0.4,0.7,0.439,LASAM
```

Figure 54 External Data – Example: Calibratable Parameters Load File

6.4 Example Calibration Definition File

This JSON file can be exported using the ngenCERF CLI “export” command and used as a template to create a new calibration job using the ngenCERF CLI “import” command.

```
{
  "metadata": {
    "source_calibration_run_id": 269,
    "source_status": "Done",
    "time_range": {
      "start_time": "2013-01-01T00:00:00+00:00",
      "end_time": "2022-12-31T22:00:00+00:00"
    },
    "job_data_dir": "/ngencertif/data/ngen-cal-work/run_calib/269_peter",
    "num_catchments": 5
  },
  "run_after_import": false,
  "gage_id": "01055000",
  "forcing_source": "AORC",
  "observational_source": "Historical",
  "geopackage_source": "Hydrofabric",
  "modules": [
    "T-Route",
    "CFE-S",
    "Noah-OWP-Modular"
  ],
  "formulation_name": "noah_cfes_troute_01055000_clone",
  "use_sloth": false,
}
```

⁵ <https://www.ogc.org/standard/geopackage/>

```

"sloth_parameters": [],
"automatic_validation": true,
"calibration_times": {
  "calibration_start_time": "2016-10-01T00:00:00Z",
  "calibration_end_time": "2017-09-01T00:00:00Z",
  "simulation_start_time": "2015-10-01T00:00:00Z",
  "simulation_end_time": "2017-09-01T00:00:00Z"
},
"validation_times": {
  "validation_start_time": "2015-10-01T00:00:00Z",
  "validation_end_time": "2016-09-01T00:00:00Z",
  "simulation_start_time": "2014-10-01T00:00:00Z",
  "simulation_end_time": "2017-09-01T00:00:00Z"
},
"streamflow_threshold": 3.88,
"peak_flow_threshold": null,
"parameters": [
  {
    "name": "b",
    "minimum": 2.0,
    "maximum": 15.0,
    "initial_value": 4.05,
    "module": "CFE-S"
  }
],
"objective_function": "KGE",
"optimization_inputs": [
  {
    "name": "r",
    "value": 0.2
  }
],
"optimization": "DDS",
"save_plot_iteration_frequency": 1,
"save_output_iteration": false,
"stop_criteria": 5
}

```

7.0 Acronyms

Acronym	Definition
CFE-S	Conceptual Functional Equivalent Model - Schaake
CFE-X	Conceptual Functional Equivalent Model - Xinanjiang
CLI	Command Line Interface
CONUS	Continental United States
COR	Pearson Correlation
CSI	Critical Success Index
EDFS	Enterprise Data and FIM Services (Formally Enterprise Hydrofabric Services) aka Data Services in this document
EVBIAS	Event Volume Bias
EWTS	Error Warning and Trapping System

FAR	False Alarm Ratio
FBIAS	Frequency bias
FIM	Flood Inundation Mapping
GUI	Graphical User Interface
HFDC	Percent bias of high flow segment of flow duration curve
KGE	Kling-Gupta Efficiency
LASAM	Lumped Arid Semi-Arid Model
LFDC	Percent bias of low flow segment of flow duration curve
LogNSE	NSE of logarithmic values
LSTM	Long Short-Term Memory Networks Model
MAE	Mean Absolute Error
MFDC	Percent bias of middle flow segment of flow duration curve
ngenCERF	ngen Calibration, Evaluation, Reforecast, Forecast
NNSE	Normalized NSE
Noah	A metaphor for modeling water and land surface processes, particularly floods, rainfall, and soil water. The Noah model was developed by the NOAA/NCEP (National Centers for Environmental Prediction) starting in the 1990s.
Noah-MP	Noah-Multi-Physics
Noah-OWP-Modular	Extended, refactored version of the Noah-MP land surface model
NSE	Nash-Sutcliffe-Efficiency
NSEwt	Weighted NSE (NSE and LogNSE)
NWM	National Water Model
OE	Operation Environment
PBIAS	Percent Bias
PDF	Postscript Document Format
PET	Potential Evapotranspiration Model
PKBIAS	Absolute Peak Flow Bias
PKTE	Peak Flow Timing Error
POD	Probability of Detection
pPKBIAS	Percent peak flow bias
PW	Parallel Works
RMSE	Root Mean Square Error
RSR	Ratio of RMSE to standard deviation of observation
Sac-SMA	Sacramento Soil Moisture Accounting Model
SCHISM	Semi-implicit Cross-scale Hydroscience Integrated System Model
SFINCS	Super-fast modelling of coastal flooding events Model
SFT	Soil Freeze Thaw Model
SLoTH	Simple Logical Tautology Handler
SMP	Soil Moisture Profiles Model
Snow-17	Snow accumulation and melt Model
TBD	To Be Determined
TopModel	A physically based, distributed watershed Model
T-Route	Tree-Based Channel Routing Model
UEB	Utah Energy Balance Model
UTC	Coordinated Universal Time